

The Lambert W Function

Real Branches, Series, Asymptotics, Bounds, Algorithms, and Applications

A self-contained mathematical introduction

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Abstract

The Lambert W function is the multivalued inverse of $w \mapsto we^w$. That definition is short, but its consequences reach from elementary equation solving to combinatorics, differential equations, circuit theory, biochemical kinetics, numerical analysis, and asymptotic inversion. This article develops the subject from first principles. Special attention is paid to the two real branches: the principal branch W_0 and the often omitted lower branch W_{-1} . We prove their domains, monotonicity, curvature, differential and integral identities, Taylor and Puiseux series, logarithmic asymptotic expansions, elementary bounds, and branch-safe iterative algorithms. We also derive representative applications rather than merely listing them.

The exposition assumes single-variable calculus and elementary power series. More advanced tools—formal residues, Lagrange inversion, and complex branch cuts—are introduced when needed and proved or explained in appendices. The attached numerical program independently checks the main expansions and produces the figures in the article. Material and perspectives from the source archive include the canonical treatment of Corless, Gonnet, Hare, Jeffrey, and Knuth; DLMF formulas; real-branch approximations of Barry and coauthors and of Iacono and Boyd; bounds for W_{-1} due to Chatzigeorgiou; continued fractions of Corcino, Corcino, and Mezo; the guaranteed iteration analysis of Loczi; and applications in electronics, kinetics, ecology, and prime-number estimates.

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How to read this article

The quickest route is Chapters 1–2, followed by Chapters 5, 6, and 7. Readers primarily interested in reliable computation can then jump to Chapter 10. Applications are collected in Chapter 13.

The notation $W(x)$ without a subscript is used only when the branch is either irrelevant or explicitly fixed by context. On the real axis, W_0 and W_{-1} are always written separately whenever two real values are possible. This convention is not cosmetic: selecting the wrong branch can change a physical time, concentration, or threshold by orders of magnitude.

Chapter 1

Definition, geometry, and the two real branches

1.1 The inverse problem

The ordinary logarithm solves the inverse problem for the exponential: $e^w = x$ if and only if $w = \log x$. The Lambert function solves the next natural inverse problem, in which the unknown occurs both outside and inside an exponential.

Definition 1.1 (Lambert W relation). For $z \in \mathbb{C}$, the values of the Lambert W relation are the complex numbers w satisfying

$$we^w = z. \quad (1.1)$$

Its single-valued branches are denoted $W_k(z)$, where $k \in \mathbb{Z}$. Thus

$$W_k(z)e^{W_k(z)} = z \quad (1.2)$$

wherever the branch W_k is defined.

The word “relation” matters over $\mathbb{C}$: just as the complex logarithm has infinitely many values, the inverse of we^w has infinitely many branches. Over $\mathbb{R}$, however, the geometry is particularly transparent.

1.2 Complete classification of real solutions

Let

$$f(w) = we^w, \quad w \in \mathbb{R}.$$

Then

$$f'(w) = e^w(1 + w), \quad f''(w) = e^w(2 + w). \quad (1.3)$$

Theorem 1.2 (Real branches). *The equation $we^w = x$ has the following real solutions.*

- (i) *If $x < -1/e$, there is no real solution.*
- (ii) *If $x = -1/e$, there is exactly one real solution, $w = -1$.*
- (iii) *If $-1/e < x < 0$, there are exactly two real solutions:*

$$-1 < W_0(x) < 0, \quad W_{-1}(x) < -1.$$

- (iv) If $x = 0$, the only finite real solution is $W_0(0) = 0$; moreover $W_{-1}(x) \rightarrow -\infty$ as $x \uparrow 0$.
(v) If $x > 0$, there is exactly one real solution, $W_0(x) > 0$.

Consequently,

$$\begin{aligned} W_0 : [-1/e, \infty) &\longrightarrow [-1, \infty), & \text{strictly increasing,} \\ W_{-1} : [-1/e, 0) &\longrightarrow (-\infty, -1], & \text{strictly decreasing.} \end{aligned}$$

Proof. The exponential is positive, so $f(w)$ has the same sign as w . By (1.3), $f'(w) < 0$ for $w < -1$, $f'(-1) = 0$, and $f'(w) > 0$ for $w > -1$. Hence f decreases on $(-\infty, -1]$ and increases on $[-1, \infty)$. Its minimum is

$$f(-1) = -e^{-1}.$$

Furthermore,

$$\lim_{w \rightarrow -\infty} we^w = 0^-, \quad f(0) = 0, \quad \lim_{w \rightarrow +\infty} we^w = +\infty.$$

The intermediate value theorem and strict monotonicity on each interval now give every assertion, including the ranges and monotonicity of the inverse branches. $\square$

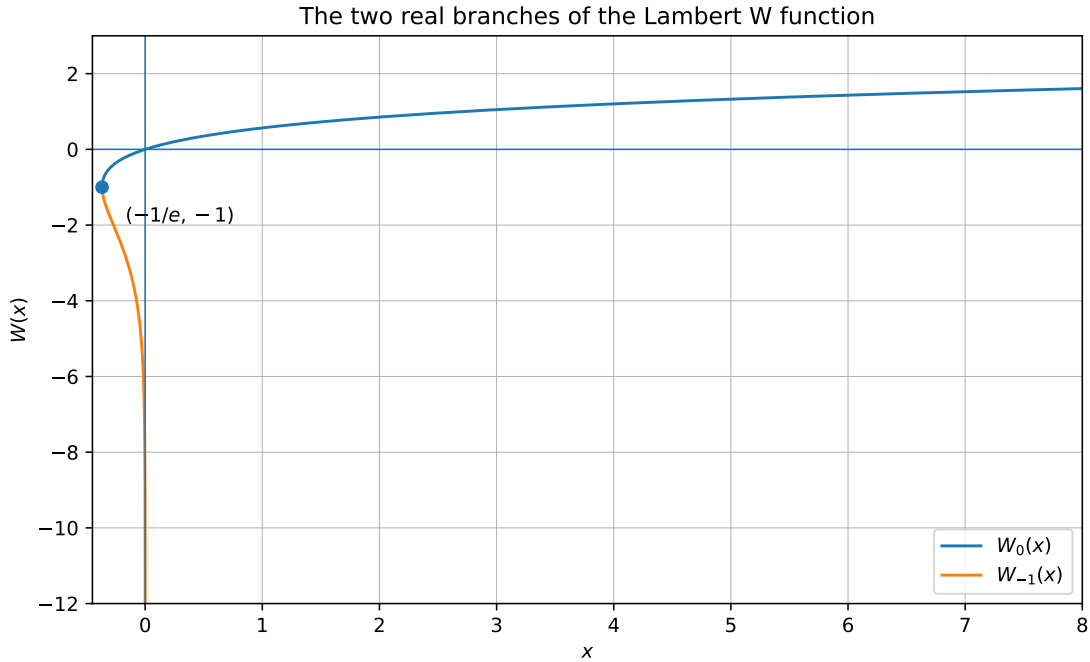


Figure 1.1: The two real branches. They meet at the square-root branch point $(-1/e, -1)$. The lower branch tends to $-\infty$ as $x \uparrow 0$.

1.3 Immediate identities and their domains

Whenever $w = W_k(z)$ is nonzero, the defining equation gives

$$e^{W_k(z)} = \frac{z}{W_k(z)}, \quad e^{-W_k(z)} = \frac{W_k(z)}{z}. \quad (1.4)$$

At $z = 0$ the first quotient has a removable singularity on W_0 because $W_0(z) \sim z$.

Taking a logarithm is more delicate over $\mathbb{C}$, but on a fixed real branch there is no ambiguity:

$$W_0(x) + \log W_0(x) = \log x \quad (x > 0), \quad (1.5)$$

$$W_j(x) + \log(-W_j(x)) = \log(-x) \quad (-1/e \leq x < 0, \ j = 0, -1). \quad (1.6)$$

Proposition 1.3 (Exact inversion of $t \mapsto te^t$). *For real t ,*

$$W_0(te^t) = t \quad \text{if } t \geq -1, \quad (1.7)$$

$$W_{-1}(te^t) = t \quad \text{if } t \leq -1. \quad (1.8)$$

In particular, for $a > 0$,

$$W_0(a \log a) = \log a \quad \text{if } a \geq e^{-1}, \quad (1.9)$$

$$W_{-1}(a \log a) = \log a \quad \text{if } 0 < a \leq e^{-1}. \quad (1.10)$$

Proof. The restrictions in (1.7) and (1.8) place t in the range of the stated inverse branch. Since t itself solves $we^w = te^t$ and the inverse is unique on that range, the formulas follow. Set $t = \log a$ for the last two identities. $\square$

Branch warning 1.4. The unqualified formula $W(te^t) = t$ is false. For example, $(-2)e^{-2} \in (-1/e, 0)$ has two preimages. One branch returns -2 and the other returns the distinct number $W_0(-2e^{-2}) \in (-1, 0)$.

1.4 Special values

Most Lambert values are transcendental-looking constants rather than familiar numbers. Exact values arise whenever the argument is deliberately written as te^t :

$$W_0(0) = 0, \quad W_0(-1/e) = W_{-1}(-1/e) = -1, \quad W_0(e) = 1, \quad W_0(2e^2) = 2, \quad W_{-1}(-3e^{-3}) = -3.$$

The positive constant

$$\Omega = W_0(1) = 0.567143290409783872999 \dots \quad (1.11)$$

is called the *omega constant*; it is the unique positive solution of $\Omega e^\Omega = 1$, equivalently $\Omega + \log \Omega = 0$.

x	$W_0(x)$	$W_{-1}(x)$ when real
$-1/e$	-1.0	-1.0
-0.1	-0.111832559158962965	-3.57715206395729722
0	0.0	$-$
1	0.567143290409783873	$-$
e	1.0	$-$
10	1.74552800274069938	$-$

Chapter 2

Putting equations into Lambert form

The main practical skill is not memorizing formulas but recognizing how to create a product ue^u in which the same transformed unknown appears in both positions.

2.1 A general affine–exponential equation

Theorem 2.1 (Affine factor times an exponential). *Let $a, c, r \neq 0$ and let b, d be constants. Every solution of*

$$(ax + b)e^{cx+d} = r \quad (2.1)$$

is of the form

$$x = \frac{1}{c} W_k \left(\frac{cr}{a} \exp \left(\frac{cb}{a} - d \right) \right) - \frac{b}{a}, \quad (2.2)$$

for a branch k on which the displayed value is defined. Conversely, every such branch value solves (2.1).

Proof. Set $u = ax + b$, so $x = (u - b)/a$. Equation (2.1) becomes

$$u \exp \left(\frac{c}{a} u + d - \frac{cb}{a} \right) = r.$$

Multiplying by $(c/a) \exp(cb/a - d)$ gives

$$\left(\frac{c}{a} u \right) \exp \left(\frac{c}{a} u \right) = \frac{cr}{a} \exp \left(\frac{cb}{a} - d \right).$$

Apply the defining inverse relation, then substitute back $u = ax + b$. Reversing the algebra proves the converse. $\square$

Example 2.2 (A decay equation with two times). Consider

$$(1 - t)e^t = q,$$

where $0 < q < 1$. Set $u = t - 1$. Then $-ue^{u+1} = q$, so

$$t = 1 + W_k(-q/e).$$

Because $-q/e \in (-1/e, 0)$, there are two real formal values. They correspond to W_0 and W_{-1} . A model that also requires $t \geq 0$ or $t \leq 1$ must use that extra information to select the physical value.

2.2 Logarithmic–linear equations

Proposition 2.3. For $A, B \neq 0$, positive solutions of

$$A \log x + Bx = C \quad (2.3)$$

are

$$x = \frac{A}{B} W_k \left(\frac{B}{A} e^{C/A} \right), \quad (2.4)$$

with exactly those branches that make the right-hand side positive and real.

Proof. Divide by A and exponentiate: $x \exp((B/A)x) = \exp(C/A)$. Multiplication by B/A creates the defining product. The converse follows by substitution. $\square$

2.3 The equation $x^x = a$

For $x > 0$, put $u = \log x$. Then $x = e^u$ and

$$x^x = a \iff x \log x = \log a \iff u e^u = \log a.$$

Hence

$$x = \exp(W_k(\log a)) = \frac{\log a}{W_k(\log a)}, \quad (2.5)$$

where the quotient is interpreted by continuity at $a = 1$.

Theorem 2.4 (Number of positive solutions of $x^x = a$). Let $a > 0$.

- (i) If $0 < a < e^{-1/e}$, there is no positive solution.
- (ii) If $a = e^{-1/e}$, the unique solution is $x = e^{-1}$.
- (iii) If $e^{-1/e} < a < 1$, there are exactly two positive solutions, obtained from $W_0(\log a)$ and $W_{-1}(\log a)$ in (2.5).
- (iv) If $a \geq 1$, there is exactly one positive solution.

Proof. The derivative of $g(x) = x \log x$ is $1 + \log x$. Thus g decreases on $(0, e^{-1})$, increases on (e^{-1}, ∞) , and has minimum $g(e^{-1}) = -1/e$. Exponentiating gives the corresponding minimum of x^x , namely $e^{-1/e}$. For $\log a \in (-1/e, 0)$, Theorem 1.2 supplies exactly two real values of W , and their exponentials are the two positive solutions. The remaining cases follow from the same monotonicity. $\square$

2.4 The equation $x^y = y^x$

For positive x, y with $x \neq 1$, the equation is equivalent to

$$\frac{\log y}{y} = \frac{\log x}{x} =: a.$$

Solving $\log y = ay$ gives

$$y = -\frac{1}{a} W_k(-a), \quad a = \frac{\log x}{x}. \quad (2.6)$$

One branch reproduces $y = x$; when $0 < a < 1/e$ the other branch gives the familiar nontrivial partner. The branch point $a = 1/e$ corresponds to $x = y = e$.

Example 2.5. Take $x = 2$. Then $a = (\log 2)/2$. The branch satisfying $W_k(-a) = -ax = -\log 2$ gives $y = 2$. The other branch yields $y = 4$, and indeed $2^4 = 4^2$.

2.5 Infinite power towers

A convergent positive tower

$$T = x^{x^{x^{\cdots}}}$$

must satisfy $T = x^T$. Therefore

$$T = -\frac{W_k(-\log x)}{\log x} = \exp(-W_k(-\log x)). \quad (2.7)$$

The equation alone does not prove that the iteration converges. At a fixed point, the derivative of $t \mapsto x^t$ is

$$x^T \log x = T \log x = \log T.$$

Thus the fixed point is locally attracting exactly when $|\log T| < 1$, or $e^{-1} < T < e$. Mapping the endpoints through $x = T^{1/T}$ produces the classical stability interval

$$e^{-e} < x < e^{1/e}.$$

At the lower part of this interval the lower real branch is essential.

Chapter 3

Differential calculus

3.1 First derivative and differential equation

Theorem 3.1 (Derivative). *On any differentiable branch and away from the branch point,*

$$W'_k(z) = \frac{W_k(z)}{z(1 + W_k(z))} = \frac{e^{-W_k(z)}}{1 + W_k(z)} = \frac{1}{z + e^{W_k(z)}}. \quad (3.1)$$

The principal branch has the removable value $W'_0(0) = 1$. Equivalently, every branch satisfies

$$z(1 + W)W' = W. \quad (3.2)$$

Proof. Differentiate $We^W = z$ implicitly:

$$e^W(1 + W)W' = 1.$$

This gives the middle expression. Substituting $e^{-W} = W/z$ gives the first; multiplying numerator and denominator of the middle expression by e^W gives the third. The Taylor expansion proved in Chapter 5 begins $W_0(z) = z - z^2 + \dots$, so $W'_0(0) = 1$. $\square$

On the real branches, the signs are immediate:

$$W'_0(x) > 0 \quad (x > -1/e), \quad W'_{-1}(x) < 0 \quad (-1/e < x < 0).$$

Both derivatives become unbounded at $x = -1/e$, with opposite signs.

3.2 Second derivative and curvature

Proposition 3.2 (Curvature). *Let $w = W_k(x)$. Then*

$$W''_k(x) = -\frac{e^{-2w}(w+2)}{(w+1)^3} = -\frac{w^2(w+2)}{x^2(w+1)^3}. \quad (3.3)$$

Consequently:

- (i) W_0 is strictly concave on $(-1/e, \infty)$.
- (ii) W_{-1} is strictly convex on $(-1/e, -2/e^2)$, has an inflection point at $x = -2/e^2$, and is strictly concave on $(-2/e^2, 0)$.

Proof. From (3.1), regard W' as the function $g(w) = e^{-w}/(1+w)$. Then

$$g'(w) = -\frac{e^{-w}(w+2)}{(w+1)^2}.$$

By the chain rule, $W'' = g'(w)W'$, giving (3.3). For W_0 , $w > -1$, so $w+2 > 0$ and the expression is negative. On W_{-1} , $w < -1$ and $(w+1)^3 < 0$. The sign changes precisely at $w = -2$, whose argument is $we^w = -2/e^2$. $\square$

The inflection of W_{-1} is an example of a property that disappears if one studies only the principal branch.

3.3 All higher derivatives

Theorem 3.3 (Derivative polynomials). *Define polynomials P_n by*

$$P_1(w) = 1, \quad P_{n+1}(w) = (1+w)P'_n(w) - (nw + 3n - 1)P_n(w). \quad (3.4)$$

Then, for $n \geq 1$,

$$\frac{d^n}{dx^n} W_k(x) = \frac{e^{-nW_k(x)} P_n(W_k(x))}{(1+W_k(x))^{2n-1}}. \quad (3.5)$$

Moreover, P_n has degree $n-1$ and leading coefficient $(-1)^{n-1}(n-1)!$.

Proof. The case $n = 1$ is (3.1). Assume (3.5). Since $d/dx = (e^{-w}/(1+w))d/dw$, differentiation gives

$$\begin{aligned} \frac{d^{n+1}W}{dx^{n+1}} &= \frac{e^{-w}}{1+w} \frac{d}{dw} \left(\frac{e^{-nw} P_n(w)}{(1+w)^{2n-1}} \right) \\ &= \frac{e^{-(n+1)w}}{(1+w)^{2n+1}} ((1+w)P'_n(w) - (nw + 3n - 1)P_n(w)), \end{aligned}$$

which is the required recurrence. If the leading coefficient of P_n is a_n , the highest-degree term in (3.4) is $-na_n w^n$. Thus $a_{n+1} = -na_n$ and $a_1 = 1$, proving the last statement. $\square$

The first few polynomials are

$$\begin{aligned} P_1(w) &= 1, \\ P_2(w) &= -(w+2), \\ P_3(w) &= 2w^2 + 8w + 9, \\ P_4(w) &= -(6w^3 + 36w^2 + 79w + 64). \end{aligned}$$

The common factor $(1+W)^{-(2n-1)}$ explains the increasing singularity of higher derivatives at the real branch point.

3.4 Differentiating compositions

For a differentiable g ,

$$\frac{d}{dx}W_k(g(x)) = \frac{W_k(g(x))}{g(x)(1 + W_k(g(x)))}g'(x). \quad (3.6)$$

A useful cancellation occurs when $g(x) = ae^{bx}$ or $g(x) = ax^r$. For example,

$$\frac{d}{dx}W_k(ae^{bx}) = b \frac{W_k(ae^{bx})}{1 + W_k(ae^{bx})}.$$

These formulas are valid only while the path $g(x)$ remains in the domain of the chosen branch.

Chapter 4

Integral calculus

4.1 The universal substitution

On any branch, set

$$w = W_k(x), \quad x = we^w, \quad dx = e^w(1 + w) dw. \quad (4.1)$$

This substitution turns many apparently exotic Lambert integrals into ordinary exponential–polynomial integrals.

Theorem 4.1 (An antiderivative of W). *On every interval or complex domain on which one branch is continuous,*

$$\int W_k(x) dx = xW_k(x) - x + e^{W_k(x)} + C. \quad (4.2)$$

Away from $x = 0$, this can also be written

$$\int W_k(x) dx = x \left(W_k(x) - 1 + \frac{1}{W_k(x)} \right) + C. \quad (4.3)$$

The form (4.2) remains nonsingular at $x = 0$ on the principal branch.

Proof. Using (4.1),

$$\int W(x) dx = \int we^w(1 + w) dw = \int (w + w^2)e^w dw.$$

Direct differentiation shows

$$\frac{d}{dw}(e^w(w^2 - w + 1)) = (w + w^2)e^w.$$

Since $x = we^w$, this antiderivative is $e^w(w^2 - w + 1) = xw - x + e^w$, proving (4.2). Formula (4.3) follows from $e^w = x/w$. $\square$

Corollary 4.2 (Three instructive definite integrals). *The real branches satisfy*

$$\int_0^e W_0(x) dx = e - 1, \quad (4.4)$$

$$\int_{-1/e}^0 W_0(x) dx = 1 - \frac{3}{e}, \quad (4.5)$$

$$\int_{-1/e}^0 W_{-1}(x) dx = -\frac{3}{e}, \quad (4.6)$$

where the last integral is improper at 0.

Proof. In terms of w , an antiderivative is $F(w) = e^w(w^2 - w + 1)$. For (4.4), the endpoints correspond to $w = 0$ and $w = 1$, giving $F(1) - F(0) = e - 1$. On W_0 , the endpoints $-1/e$ and 0 correspond to $w = -1$ and $w = 0$, giving $F(0) - F(-1) = 1 - 3/e$. On W_{-1} , they correspond to $w = -1$ and $w \rightarrow -\infty$. Since $F(w) \rightarrow 0$ as $w \rightarrow -\infty$, the improper integral is $0 - F(-1) = -3/e$. $\square$

It is striking that the unbounded lower branch is nevertheless integrable at 0 . Later we shall prove $W_{-1}(x) \sim \log(-x)$ there, and $\int_{-\varepsilon}^0 |\log(-x)| dx$ is finite.

4.2 A family of polynomial–Lambert integrals

For an integer $n \geq 0$ and a nonzero constant a , define

$$Q_{n,a}(w) = \sum_{j=0}^n \frac{(-1)^j n!}{(n-j)! a^{j+1}} w^{n-j}. \quad (4.7)$$

Repeated integration by parts, or one differentiation, gives

$$\frac{d}{dw} (e^{aw} Q_{n,a}(w)) = w^n e^{aw}. \quad (4.8)$$

Theorem 4.3 (Integrals of $x^p W(x)^q$). *Let p, q be nonnegative integers and put $N = p + q$, $a = p + 1$. Then*

$$\int x^p W_k(x)^q dx = e^{aW_k(x)} (Q_{N,a}(W_k(x)) + Q_{N+1,a}(W_k(x))) + C. \quad (4.9)$$

The formula is valid on any branch interval on which the powers are defined.

Proof. Substitution (4.1) gives

$$\begin{aligned} x^p W(x)^q dx &= (we^w)^p w^q e^w (1+w) dw \\ &= (w^N + w^{N+1}) e^{(p+1)w} dw. \end{aligned}$$

Apply (4.8) to the two terms. $\square$

Example 4.4. For $q = 2, p = 0$, the theorem gives

$$\int W(x)^2 dx = e^{W(x)} (W(x)^3 - 2W(x)^2 + 4W(x) - 4) + C.$$

Differentiation is a useful check. The same expression applies to W_0 and W_{-1} ; only the branch substitution differs.

4.3 The logarithmic measure dx/x

When $x \neq 0$, (4.1) yields

$$\frac{dx}{x} = \frac{1+w}{w} dw. \quad (4.10)$$

Thus, for an integer $q \geq 1$,

$$\int \frac{W_k(x)^q}{x} dx = \frac{W_k(x)^q}{q} + \frac{W_k(x)^{q+1}}{q+1} + C. \quad (4.11)$$

The case $q = 0$ recovers $\log|x| = W + \log|W|$ on the real branches.

Chapter 5

Taylor series, Lagrange inversion, and rooted trees

5.1 Lagrange–Bürmann inversion

We use the following coefficient form of the inversion theorem. A proof by formal residues is given in Appendix A; it requires no analytic convergence assumptions and therefore applies first as an identity of formal power series.

Theorem 5.1 (Lagrange–Bürmann). *Suppose $w = w(z)$ is the unique formal power series with $w(0) = 0$ satisfying*

$$w = z\phi(w), \quad \phi(0) \neq 0. \quad (5.1)$$

Then, for every formal series H and every $n \geq 1$,

$$[z^n]H(w(z)) = \frac{1}{n}[t^{n-1}]H'(t)\phi(t)^n. \quad (5.2)$$

For Lambert W , the defining equation is equivalent to

$$W_0(z) = ze^{-W_0(z)}, \quad (5.3)$$

so $\phi(t) = e^{-t}$.

5.2 The Taylor series at the origin

Theorem 5.2 (Lambert’s series). *For $|z| < 1/e$,*

$$W_0(z) = \sum_{n=1}^{\infty} \frac{(-n)^{n-1}}{n!} z^n = z - z^2 + \frac{3}{2}z^3 - \frac{8}{3}z^4 + \frac{125}{24}z^5 - \dots. \quad (5.4)$$

Its radius of convergence is exactly $1/e$.

Proof. Apply Theorem 5.1 with $H(t) = t$ and $\phi(t) = e^{-t}$:

$$[z^n]W_0(z) = \frac{1}{n}[t^{n-1}]e^{-nt} = \frac{1}{n} \frac{(-n)^{n-1}}{(n-1)!} = \frac{(-n)^{n-1}}{n!}.$$

Let $a_n = n^{n-1}/n!$ be the absolute coefficient. Then

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^n}{(n+1)!} \frac{n!}{n^{n-1}} = \left(1 + \frac{1}{n}\right)^{n-1} \rightarrow e.$$

The ratio test therefore gives radius $1/e$. The singular branch point at $z = -1/e$, independently derived in Chapter 6, confirms that analytic continuation cannot enlarge the Taylor disk. $\square$

Corollary 5.3. *For $n \geq 1$,*

$$W_0^{(n)}(0) = (-n)^{n-1}. \quad (5.5)$$

The series converges at both real boundary points $z = \pm 1/e$. Indeed, Stirling's formula gives

$$\frac{n^{n-1}}{n!e^n} \sim \frac{1}{\sqrt{2\pi} n^{3/2}},$$

so absolute convergence holds. At $z = -1/e$ the sum is -1 by continuity.

Branch warning 5.4. There is no Taylor series for W_{-1} at 0. The lower branch diverges to $-\infty$ there and has a logarithmic singularity. Its natural local expansion is the asymptotic series of Chapter 7, not a power series in x .

5.3 Powers and exponentials of W_0

Theorem 5.5 (Powers of the principal branch). *For an integer $r \geq 1$ and $|z| < 1/e$,*

$$W_0(z)^r = \sum_{n=r}^{\infty} \frac{-r(-n)^{n-r-1}}{(n-r)!} z^n. \quad (5.6)$$

More generally, for $\alpha \in \mathbb{C}$,

$$\left(\frac{W_0(z)}{z}\right)^\alpha = e^{-\alpha W_0(z)} = \sum_{n=0}^{\infty} \frac{\alpha(n+\alpha)^{n-1}}{n!} (-z)^n, \quad (5.7)$$

with the $n = 0$ coefficient interpreted as 1.

Proof. For (5.6), use Theorem 5.1 with $H(t) = t^r$:

$$\begin{aligned} [z^n]W_0(z)^r &= \frac{r}{n} [t^{n-r}]e^{-nt} \\ &= \frac{r}{n} \frac{(-n)^{n-r}}{(n-r)!} = \frac{-r(-n)^{n-r-1}}{(n-r)!}. \end{aligned}$$

For (5.7), the defining equation gives $W_0(z)/z = e^{-W_0(z)}$. Apply Lagrange–Bürmann to $H(t) = e^{-\alpha t}$:

$$\begin{aligned} [z^n]e^{-\alpha W_0(z)} &= \frac{1}{n} [t^{n-1}](-\alpha)e^{-(n+\alpha)t} \\ &= \frac{\alpha(-1)^n(n+\alpha)^{n-1}}{n!} \quad (n \geq 1). \end{aligned}$$

The constant term is 1. $\square$

5.4 The tree function

Define

$$T(z) = -W_0(-z). \quad (5.8)$$

Then

$$T(z) = \sum_{n=1}^{\infty} \frac{n^{n-1}}{n!} z^n, \quad T(z)e^{-T(z)} = z, \quad T(z) = ze^{T(z)}. \quad (5.9)$$

Cayley's theorem says that there are n^{n-2} unrooted labeled trees on n vertices and therefore $n \cdot n^{n-2} = n^{n-1}$ rooted labeled trees. Thus T is the exponential generating function of rooted labeled trees.

The functional equation $T = ze^T$ also has a direct combinatorial meaning. A rooted labeled tree consists of one labeled root together with an unordered set of rooted labeled subtrees attached to it. In exponential generating functions, a labeled root contributes z and an unordered set contributes the exponential. The analytic inverse relation and the combinatorial recursive construction are therefore the same equation viewed from two directions.

5.5 Expansion about an ordinary point

Let $x_0 = w_0 e^{w_0}$ with $w_0 \neq -1$. The inverse function theorem gives an ordinary Taylor expansion of the branch through (x_0, w_0) . Its first terms are

$$\begin{aligned} W(x_0 + h) = w_0 + \frac{e^{-w_0}}{1 + w_0} h - \frac{e^{-2w_0}(w_0 + 2)}{2(1 + w_0)^3} h^2 \\ + \frac{e^{-3w_0}(2w_0^2 + 8w_0 + 9)}{6(1 + w_0)^5} h^3 + \mathcal{O}(h^4). \end{aligned} \quad (5.10)$$

This applies equally to W_0 and W_{-1} at every interior point of their real domains. The failure at $w_0 = -1$ is exactly why the branch-point expansion requires half-integer powers.

Chapter 6

The square-root branch point at $-1/e$

6.1 Why an ordinary Taylor series fails

The inverse function theorem fails at $w = -1$ because $f'(-1) = 0$. The first nonzero term of $f(-1+u) - f(-1)$ is quadratic, so the inverse naturally contains a square root.

Put

$$x = -\frac{1}{e} + h, \quad w = -1 + u, \quad p = \sqrt{2(1+ex)} = \sqrt{2eh}. \quad (6.1)$$

Then

$$\begin{aligned} 1+ex &= 1 + (u-1)e^u \\ &= \frac{u^2}{2} + \frac{u^3}{3} + \frac{u^4}{8} + \frac{u^5}{30} + \frac{u^6}{144} + \cdots \\ &= \sum_{n=2}^{\infty} \frac{n-1}{n!} u^n. \end{aligned} \quad (6.2)$$

Thus $p = u + \mathcal{O}(u^2)$ on the upper side and $p = -u + \mathcal{O}(u^2)$ on the lower side.

6.2 A single analytic inverse for both real branches

Define the analytic function near $u = 0$ by

$$G(u) = u \sqrt{\frac{2(1+(u-1)e^u)}{u^2}}, \quad G(0) = 0. \quad (6.3)$$

The quotient under the square root tends to 1, so the square root is the analytic one with value 1 at the origin. Hence $G'(0) = 1$, and G has a local analytic inverse U with $U(0) = 0$ and $U'(0) = 1$.

Theorem 6.1 (Puiseux expansion of both real branches). *Let $p = \sqrt{2(1+ex)}$. As $x \downarrow -1/e$,*

$$W_0(x) = -1 + U(p), \quad (6.4)$$

$$W_{-1}(x) = -1 + U(-p), \quad (6.5)$$

where

$$U(q) = q - \frac{1}{3}q^2 + \frac{11}{72}q^3 - \frac{43}{540}q^4 + \frac{769}{17280}q^5 - \frac{221}{8505}q^6 + \mathcal{O}(q^7). \quad (6.6)$$

More generally, if $U(q) = \sum_{n \geq 1} c_n q^n$, then

$$c_n = \frac{1}{n} [t^{n-1}] \left(\frac{t}{G(t)} \right)^n. \quad (6.7)$$

Proof. Equation (6.2) says $G(u)^2 = p^2$. On the principal branch, $u = W_0(x) + 1 \geq 0$, so $G(u) = p$ near the branch point and $u = U(p)$. On the lower branch, $u = W_{-1}(x) + 1 \leq 0$, so $G(u) = -p$ and $u = U(-p)$. This proves (6.4)–(6.5).

For the coefficients, rewrite $q = G(u)$ as $u = q u / G(u)$ and apply Theorem 5.1. Formula (6.7) follows. Expanding the analytic factor $t/G(t)$ and matching powers gives

$$c_1 = 1, \quad c_2 = -\frac{1}{3}, \quad c_3 = \frac{11}{72}, \quad c_4 = -\frac{43}{540}, \quad c_5 = \frac{769}{17280}, \quad c_6 = -\frac{221}{8505}.$$

Substitution into (6.2) verifies these coefficients directly. $\square$

Writing the two branches explicitly,

$$W_0(x) = -1 + p - \frac{1}{3}p^2 + \frac{11}{72}p^3 - \frac{43}{540}p^4 + \frac{769}{17280}p^5 - \frac{221}{8505}p^6 + \cdots, \quad (6.8)$$

$$W_{-1}(x) = -1 - p - \frac{1}{3}p^2 - \frac{11}{72}p^3 - \frac{43}{540}p^4 - \frac{769}{17280}p^5 - \frac{221}{8505}p^6 + \cdots. \quad (6.9)$$

The even powers agree and the odd powers have opposite signs. This is not a coincidence but the exact symmetry $U(p)$ versus $U(-p)$.

Corollary 6.2 (Branch separation and mean). *As $p \rightarrow 0^+$,*

$$W_0(x) - W_{-1}(x) = 2p + \frac{11}{36}p^3 + \frac{769}{8640}p^5 + \mathcal{O}(p^7), \quad (6.10)$$

$$\frac{W_0(x) + W_{-1}(x)}{2} = -1 - \frac{1}{3}p^2 - \frac{43}{540}p^4 - \frac{221}{8505}p^6 + \mathcal{O}(p^8). \quad (6.11)$$

Proof. Subtract and add (6.8) and (6.9). $\square$

6.3 Vertical tangents

Since $p = \sqrt{2eh}$ and $dp/dx = e/p$, the first term of the Puiseux series gives

$$W'_0(x) \sim \frac{e}{p} = \sqrt{\frac{e}{2(x+1/e)}}, \quad (6.12)$$

$$W'_{-1}(x) \sim -\frac{e}{p} = -\sqrt{\frac{e}{2(x+1/e)}}. \quad (6.13)$$

Thus the two graphs meet continuously but with opposite vertical tangents. Higher derivatives diverge with increasing half-integer powers, consistent with (3.5).

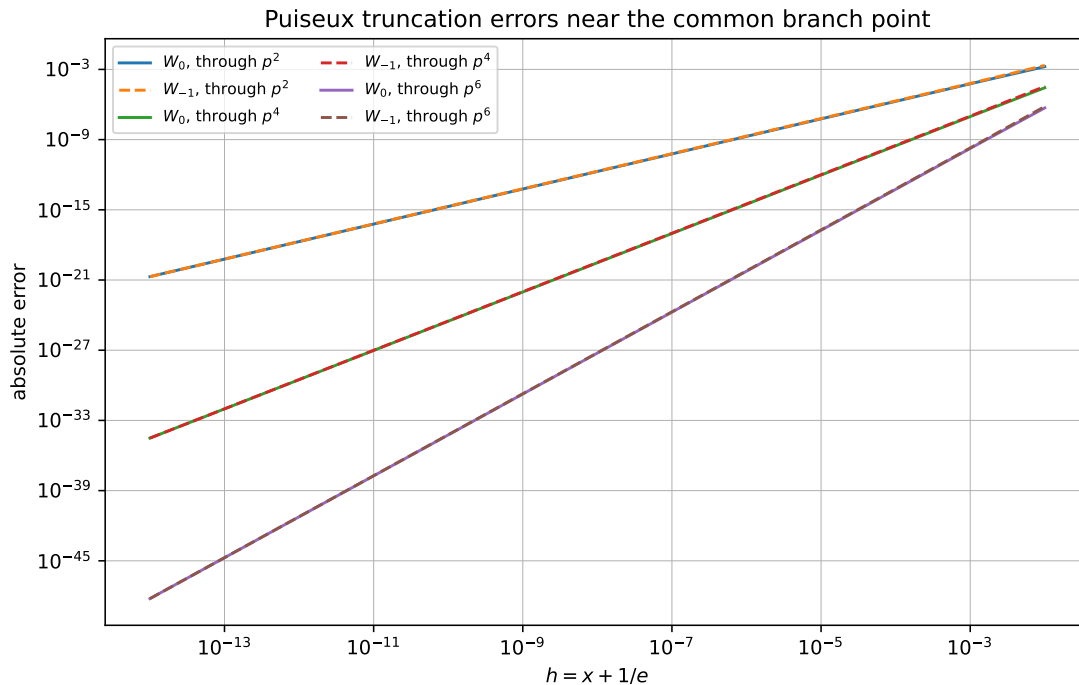


Figure 6.1: Absolute errors of Puiseux truncations on both branches. Because the same analytic inverse U controls both branches, their errors have nearly identical orders.

6.4 A practical branch-point approximation

For x close to $-1/e$, the variable p is vastly better conditioned than $x + 1/e$ itself. Evaluating (6.8) or (6.9) through p^5 or p^6 supplies an excellent initial value for an iterative method. It also automatically selects the branch: use $+p$ for W_0 and $-p$ for W_{-1} .

Care is needed in floating-point arithmetic when x is supplied as $-1/e + h$ with tiny h : computing $1 + ex$ by naive subtraction may lose most significant digits. If h is known separately, compute $p = \sqrt{2eh}$ directly; otherwise use extended precision or a fused multiply-add operation where available.

Chapter 7

Logarithmic asymptotic expansions

7.1 Unsigned Stirling numbers of the first kind

The unsigned Stirling number $\begin{bmatrix} n \\ j \end{bmatrix}$ counts permutations of n objects with exactly j cycles. Algebraically it is characterized by

$$t(t+1)\cdots(t+n-1) = \sum_{j=0}^n \begin{bmatrix} n \\ j \end{bmatrix} t^j. \quad (7.1)$$

We shall use the standard logarithmic coefficient identity

$$[u^n](\log(1-u))^j = (-1)^j \frac{j!}{n!} \begin{bmatrix} n \\ j \end{bmatrix} \quad (n \geq j \geq 1). \quad (7.2)$$

It follows by substituting $-u$ into the exponential generating function for signed Stirling numbers, or by comparing powers in $(1-u)^t = \exp(t \log(1-u))$.

7.2 The principal branch at positive infinity

Put

$$L_1 = \log x, \quad L_2 = \log \log x. \quad (7.3)$$

The elementary first approximation is $W_0(x) \approx L_1$, because $W_0 e^{W_0} = x$. Taking a logarithm shows the correction immediately: $W_0 + \log W_0 = L_1$, so $W_0 \approx L_1 - L_2$.

Theorem 7.1 (Complete logarithmic expansion of W_0). *For every fixed $N \geq 1$, as $x \rightarrow +\infty$,*

$$\begin{aligned} W_0(x) = L_1 - L_2 + \sum_{\ell=1}^N \frac{1}{L_1^\ell} \sum_{m=1}^{\ell} \frac{(-1)^{\ell-m}}{m!} \begin{bmatrix} \ell \\ \ell-m+1 \end{bmatrix} L_2^m \\ + \mathcal{O}\left(\frac{L_2^{N+1}}{L_1^{N+1}}\right). \end{aligned} \quad (7.4)$$

The first terms are

$$W_0(x) = L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2(-2+L_2)}{2L_1^2} + \frac{L_2(6-9L_2+2L_2^2)}{6L_1^3}$$

$$\begin{aligned}
& + \frac{L_2(-12 + 36L_2 - 22L_2^2 + 3L_2^3)}{12L_1^4} \\
& + \frac{L_2(60 - 300L_2 + 350L_2^2 - 125L_2^3 + 12L_2^4)}{60L_1^5} + \dots.
\end{aligned} \tag{7.5}$$

Proof. Write $t = 1/L_1$, $B = L_2$, and set

$$W_0(x) = L_1(1 - y).$$

The equation $W_0 + \log W_0 = L_1$ becomes

$$y = t(B + \log(1 - y)). \tag{7.6}$$

Treating B as a parameter, Lagrange inversion gives

$$y = \sum_{n \geq 1} \frac{t^n}{n} [u^{n-1}](B + \log(1 - u))^n.$$

Since $W_0 = L_1 - y/t$, the coefficient of $t^\ell = L_1^{-\ell}$ for $\ell \geq 1$ is

$$-\frac{1}{\ell + 1} [u^\ell](B + \log(1 - u))^{\ell+1}. \tag{7.7}$$

Expand by the binomial theorem and apply (7.2). If j is the power of the logarithm and $m = \ell + 1 - j$, expression (7.7) simplifies to

$$\sum_{m=1}^{\ell} \frac{(-1)^{\ell-m}}{m!} \left[\begin{matrix} \ell \\ \ell - m + 1 \end{matrix} \right] B^m,$$

which proves the coefficient formula. Finally, $tB = L_2/L_1 \rightarrow 0$. The analytic implicit-function theorem applied to (7.6) therefore makes the finite formal expansion asymptotic, with the stated order of remainder. $\square$

In particular,

$$W_0(x) = \log x - \log \log x + o(1). \tag{7.8}$$

The error actually tends to zero, not merely to a bounded quantity.

7.3 The lower branch near zero

For $x \in (-1/e, 0)$, set

$$\eta = \log \frac{1}{-x} = -\log(-x), \quad B = \log \eta, \quad u = -W_{-1}(x) > 1. \tag{7.9}$$

Since $-x = ue^{-u}$,

$$u - \log u = \eta. \tag{7.10}$$

This is the real lower-branch analogue of $W_0 + \log W_0 = L_1$.

Theorem 7.2 (Complete real expansion of W_{-1}). *For every fixed $N \geq 1$, as $x \uparrow 0$,*

$$\begin{aligned} -W_{-1}(x) &= \eta + B + \sum_{\ell=1}^N \frac{1}{\eta^\ell} \sum_{m=1}^{\ell} \frac{(-1)^{m-1}}{m!} \begin{bmatrix} \ell \\ \ell - m + 1 \end{bmatrix} B^m \\ &\quad + \mathcal{O}\left(\frac{B^{N+1}}{\eta^{N+1}}\right). \end{aligned} \quad (7.11)$$

In particular,

$$\begin{aligned} W_{-1}(x) &= -\left(\eta + B + \frac{B}{\eta} + \frac{B(2-B)}{2\eta^2} + \frac{B(6-9B+2B^2)}{6\eta^3} \right. \\ &\quad + \frac{B(12-36B+22B^2-3B^3)}{12\eta^4} \\ &\quad \left. + \frac{B(60-300B+350B^2-125B^3+12B^4)}{60\eta^5} + \dots \right). \end{aligned} \quad (7.12)$$

Proof. Set $u = \eta(1+y)$ and $t = 1/\eta$. Equation (7.10) becomes

$$y = t(B + \log(1+y)). \quad (7.13)$$

Lagrange inversion yields

$$y = \sum_{n \geq 1} \frac{t^n}{n} [s^{n-1}](B + \log(1+s))^n.$$

Now

$$[s^\ell](\log(1+s))^j = (-1)^{\ell-j} \frac{j!}{\ell!} \begin{bmatrix} \ell \\ j \end{bmatrix}.$$

Since $u = \eta + y/t$, the same binomial simplification as in the preceding proof gives the inner sum in (7.11). Because $B/\eta \rightarrow 0$, the analytic implicit-function argument again supplies the remainder estimate. $\square$

The first two terms alone give the memorable rule

$$W_{-1}(x) = \log(-x) - \log(-\log(-x)) + o(1) \quad (x \uparrow 0). \quad (7.14)$$

It looks formally like the W_0 expansion, but $\log(-x)$ is large and negative, so the branch tends to $-\infty$.

7.4 Derivatives in the asymptotic regimes

From (3.1),

$$W_0'(x) = \frac{1}{x} \left(1 - \frac{1}{W_0(x)} + \mathcal{O}(W_0(x)^{-2}) \right) \quad (x \rightarrow \infty), \quad (7.15)$$

$$W_{-1}'(x) = \frac{1}{x} \left(1 - \frac{1}{W_{-1}(x)} + \mathcal{O}(W_{-1}(x)^{-2}) \right) \quad (x \uparrow 0). \quad (7.16)$$

The second derivative formula then gives $W_0''(x) \sim -1/x^2$ and $W_{-1}''(x) \sim -1/x^2$ in their respective limits. For W_{-1} , $x < 0$, so $W_{-1}'(x) \sim 1/x < 0$.

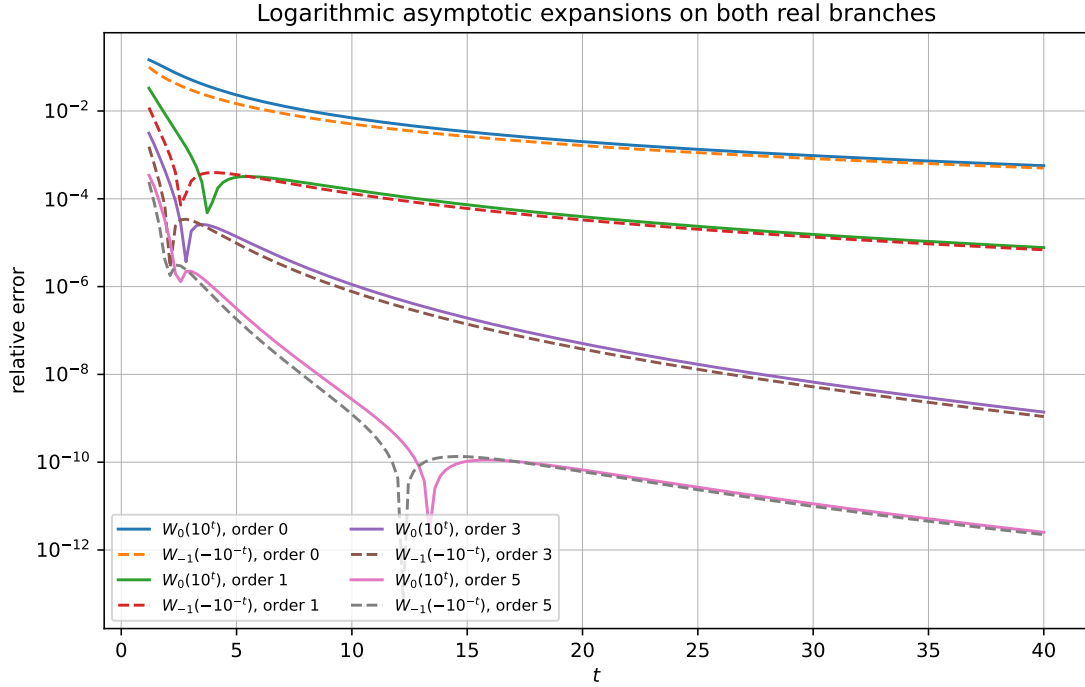


Figure 7.1: Relative errors of paired logarithmic expansions. Solid curves use $W_0(10^t)$; dashed curves use $W_{-1}(-10^{-t})$. Each additional pair of terms markedly extends the useful range.

7.5 Other complex branches

Choose a branch of the logarithm and define

$$L_{1,k} = \text{Log } z + 2\pi i k, \quad L_{2,k} = \text{Log } L_{1,k}. \quad (7.17)$$

Away from the relevant cuts and Stokes transitions, the same formal expansion as (7.4) gives

$$W_k(z) \sim L_{1,k} - L_{2,k} + \sum_{\ell \geq 1} \frac{1}{L_{1,k}^\ell} \sum_{m=1}^{\ell} \frac{(-1)^{\ell-m}}{m!} \left[\begin{matrix} \ell \\ \ell - m + 1 \end{matrix} \right] L_{2,k}^m. \quad (7.18)$$

The integer k enters first through the $2\pi i k$ in $L_{1,k}$. This explains why the branches are approximately vertically separated by $2\pi i$ for large $|k|$ or large $|z|$.

Chapter 8

Bounds and inequalities

Asymptotic formulas describe limiting behavior, but an inequality gives a certificate at a finite argument. This chapter begins with bounds that need only elementary calculus and then develops sharper estimates adapted to the lower branch.

8.1 Two global bounds for the positive principal branch

Theorem 8.1. *For every $x \geq 0$,*

$$\frac{x}{1+x} \leq W_0(x) \leq x. \quad (8.1)$$

Both inequalities are strict when $x > 0$.

Proof. Put $w = W_0(x) \geq 0$, so that $x = we^w$. Since $e^w \geq 1$, $x \geq w$, proving the upper bound. For the lower bound, after cancelling $w > 0$, the desired inequality is equivalent to

$$\frac{e^w}{1+we^w} < 1, \quad \text{or} \quad 1 + (w-1)e^w > 0.$$

The last function is zero at $w = 0$ and has derivative $we^w > 0$ for $w > 0$. Hence it is positive there. At $x = 0$ all three quantities in (8.1) vanish. $\square$

For small x , the upper bound has the correct leading term. For large x , logarithmic bounds are much better.

Theorem 8.2 (nested-logarithm bounds). *Let $x > e$, put $A = \log x$, and define*

$$a_0 = A, \quad a_{n+1} = A - \log a_n. \quad (8.2)$$

Then every a_n is positive and

$$a_1 < a_3 < \cdots < W_0(x) < \cdots < a_4 < a_2 < a_0. \quad (8.3)$$

In particular,

$$\log x - \log \log x < W_0(x) < \log x - \log(\log x - \log \log x). \quad (8.4)$$

Proof. Write $w = W_0(x)$. Taking logarithms of $we^w = x$ gives $w = A - \log w$. Thus w is a fixed point of the decreasing function $g(t) = A - \log t$. Since $A = w + \log w > w > 1$, we have $a_0 > w$ and hence $a_1 = g(a_0) < g(w) = w$. Conversely $a_2 = g(a_1) > w$. Also $A - \log A > 1$ for $A > 1$, so the whole interval $[a_1, a_0]$ is positive and is mapped into itself.

The map $g \circ g$ is increasing. From $a_2 < a_0$ it follows inductively that $a_{2n+2} < a_{2n}$; applying the decreasing map g then shows $a_{2n+3} > a_{2n+1}$. The even and odd subsequences are therefore nested around w , proving (8.3). The first odd and first nontrivial even bounds give (8.4). $\square$

The iteration in (8.2) is only linearly convergent, but its alternation is convenient when a simple symbolic upper and lower bound is preferable to a high-precision number.

8.2 Bounds for W_0 on the negative interval

Theorem 8.3. For $-1/e < x < 0$,

$$-1 + \sqrt{1 + ex} < W_0(x) < x. \quad (8.5)$$

Proof. Let $w = W_0(x) \in (-1, 0)$. Because $e^w < 1$ and $w < 0$, $x = we^w > w$, which is the upper bound.

For the lower bound put $t = 1 + w \in (0, 1)$. Since $x = (t - 1)e^{t-1}$, the desired inequality is equivalent to

$$1 + (t - 1)e^t < t^2.$$

Define $q(t) = t^2 - 1 - (t - 1)e^t$. Then $q(0) = q(1) = 0$ and

$$q'(t) = t(2 - e^t).$$

Thus q increases on $(0, \log 2)$ and decreases on $(\log 2, 1)$. It is therefore positive throughout $(0, 1)$, proving the result. $\square$

The square root in the lower bound is not accidental: it detects the branch point. Its coefficient is not optimal—the leading Puiseux term uses $\sqrt{2(1 + ex)}$ —but the bound remains valid over the entire interval.

8.3 Elementary logarithmic bounds for W_{-1}

Theorem 8.4. For $-1/e \leq x < 0$,

$$\frac{e}{e-1} \log(-x) \leq W_{-1}(x) \leq \log(-x) - \log(-\log(-x)). \quad (8.6)$$

Equality in the right inequality occurs at $x = -1/e$; equality in the left inequality occurs when $W_{-1}(x) = -e$.

Proof. Set

$$u = -W_{-1}(x) \geq 1, \quad \eta = -\log(-x) = u - \log u \geq 1.$$

The right inequality in (8.6) is equivalent to $u \geq \eta + \log \eta$. But $u \geq \eta$ and therefore $u - \eta = \log u \geq \log \eta$.

For the left inequality, put $c = e/(e - 1)$. It is equivalent to $u \leq c\eta$. After multiplying by $e - 1$, this becomes

$$u - e \log u \geq 0.$$

The function on the left has derivative $1 - e/u$, decreases on $[1, e]$, increases on $[e, \infty)$, and has minimum zero at $u = e$. $\square$

The upper bound is exactly the first two terms of the logarithmic asymptotic expansion. The lower bound is coarser near zero, but it is global and has a short proof.

8.4 Square-root bounds for the lower branch

The following result, adapted from Chatzigeorgiou's analysis [8], is especially useful near $-1/e$.

Lemma 8.5. *For $v \geq 0$,*

$$\log(1+v) \leq v - \frac{v^2}{2(1+v/3)^2}. \quad (8.7)$$

Also $\log(1+v) \geq v - v^2/2$.

Proof. For the first inequality let

$$h(v) = \log(1+v) - v + \frac{v^2}{2(1+v/3)^2}.$$

A direct differentiation gives

$$h'(v) = -\frac{v^3(v+9)}{(v+1)(v+3)^3} \leq 0.$$

Since $h(0) = 0$, we have $h(v) \leq 0$. For the second inequality, define $k(v) = \log(1+v) - v + v^2/2$. Then $k'(v) = v^2/(1+v) \geq 0$ and $k(0) = 0$. $\square$

Theorem 8.6 (two-sided branch-point bounds). *For $s > 0$,*

$$-1 - \sqrt{2s} - s < W_{-1}(-e^{-1-s}) < -1 - \sqrt{2s} - \frac{2}{3}s. \quad (8.8)$$

Proof. Write

$$W_{-1}(-e^{-1-s}) = -1 - v, \quad v > 0.$$

Substitution into $We^W = -e^{-1-s}$ gives

$$s = g(v) := v - \log(1+v). \quad (8.9)$$

For a real constant c , define

$$F_c(v) = cg(v) + \sqrt{2g(v)} - v.$$

Because $g'(v) = v/(1+v)$,

$$F'_c(v) = \frac{v - (1+v(1-c))\sqrt{2g(v)}}{(1+v)\sqrt{2g(v)}}. \quad (8.10)$$

When $c = 2/3$, inequality (8.7) says $g(v) \geq v^2/[2(1+v/3)^2]$; hence (8.10) is nonpositive. Since $F_{2/3}(0) = 0$, we get $F_{2/3}(v) < 0$ for $v > 0$. When $c = 1$, the second part of Theorem 8.5 says $g(v) \leq v^2/2$, so $F'_1(v) \geq 0$ and $F_1(v) > 0$ for $v > 0$. Therefore

$$\frac{2}{3}s < v - \sqrt{2s} < s.$$

Equivalently, $\sqrt{2s} + 2s/3 < v < \sqrt{2s} + s$. Replacing v by $-1 - W_{-1}(-e^{-1-s})$ reverses the signs and yields (8.8). $\square$

The same parameter $s = -1 - \log(-x)$ will provide a guaranteed starting value for W_{-1} in Chapter 10.

8.5 Transporting bounds through monotone functions

A bound on W immediately produces many others. For example, $e^{W_0(x)} = x/W_0(x)$, so positive lower and upper bounds $L(x) < W_0(x) < U(x)$ give

$$\frac{x}{U(x)} < e^{W_0(x)} < \frac{x}{L(x)}.$$

If a solution is $y = A + BW_k(C)$, the order is preserved when $B > 0$ and reversed when $B < 0$. This elementary observation is valuable in applications: one should propagate the inequality only after checking both the branch and the sign of every prefactor.

Chapter 9

Integral, contour, and continued-fraction representations

A representation is useful when it exposes structure that the defining implicit equation hides. Some representations are naturally principal-branch objects; the failure to extend them to W_{-1} is itself informative.

9.1 The Wright omega function

For real t , define the Wright omega function by

$$\omega(t) = W_0(e^t). \quad (9.1)$$

It is the unique positive solution of

$$\omega + \log \omega = t. \quad (9.2)$$

Conversely, $W_0(x) = \omega(\log x)$ for $x > 0$. Differentiation gives

$$\omega'(t) = \frac{\omega(t)}{1 + \omega(t)}. \quad (9.3)$$

This form avoids constructing the possibly enormous number e^t and is therefore often preferable in floating-point computation. In the complex plane the standard Wright omega function incorporates an “unwinding number” that selects the appropriate logarithm; see Corless and Jeffrey for the full branch convention.

For the lower real branch it is more natural to use $u = -W_{-1}(-e^{-\eta})$, which satisfies $u - \log u = \eta$. The change of sign in front of the logarithm is precisely what distinguishes the lower-branch asymptotic expansion from (9.2).

9.2 A contour formula for any simple branch value

Theorem 9.1. *Let $z \in \mathbb{C}$, and let Γ be a positively oriented simple closed contour in the w -plane. Suppose $we^w - z$ has exactly one zero w_k inside Γ , that zero is simple, and there are no zeros on Γ . Then*

$$w_k = \frac{1}{2\pi i} \int_{\Gamma} \frac{w e^w (1+w)}{w e^w - z} dw. \quad (9.4)$$

Thus, when the contour isolates the k th inverse value, the right side equals $W_k(z)$.

Proof. The integrand is $wf'(w)/(f(w) - z)$ with $f(w) = we^w$. At a simple zero w_k , its residue is

$$\frac{w_k f'(w_k)}{f'(w_k)} = w_k.$$

There are no other poles inside Γ , so the residue theorem gives (9.4). $\square$

At $z = -1/e$ the zero $w = -1$ is double because $f'(-1) = 0$; the hypotheses then fail. This is the contour-theoretic reflection of the square-root branch point.

9.3 A Fourier–logarithm integral for W_0

The following representation is a transparent consequence of the Taylor series and Fourier orthogonality. Corcino, Corcino, and Mezo extend the same integral, with a careful continuous choice of logarithm, to the real interval $[-1/e, e]$ [11].

Theorem 9.2. For real $|x| < 1/e$,

$$W_0(x) = \frac{1}{\pi} \operatorname{Re} \int_0^\pi \log \left(\frac{\exp(e^{it}) - xe^{-it}}{\exp(e^{it}) - xe^{it}} \right) dt, \quad (9.5)$$

where the logarithm is obtained by its power series at $x = 0$.

Proof. Let $A(t) = \exp(e^{it})$. Since $|xe^{\pm it}/A(t)| \leq e|x| < 1$, both logarithms may be expanded uniformly and integrated term by term. Their difference is

$$\log \frac{A - xe^{-it}}{A - xe^{it}} = 2i \sum_{n \geq 1} \frac{x^n}{n} e^{-ne^{it}} \sin(nt).$$

Expand once more:

$$e^{-ne^{it}} = \sum_{j \geq 0} \frac{(-n)^j}{j!} e^{ijt}.$$

Because $\operatorname{Re}(ie^{ijt}) = -\sin(jt)$ and

$$\int_0^\pi \sin(jt) \sin(nt) dt = \begin{cases} \pi/2, & j = n, \\ 0, & j \neq n, \end{cases}$$

the coefficient of x^n on the right of (9.5) is

$$-\frac{(-n)^n}{n n!} = \frac{(-n)^{n-1}}{n!}.$$

These are exactly the Taylor coefficients in (5.4), proving the identity. $\square$

There is no analogous Taylor-based integral centered at zero for W_{-1} : that branch diverges logarithmically as $x \uparrow 0$. A lower-branch integral must instead be based on a branch-point or logarithmic coordinate.

9.4 Euler's conversion of a series to a continued fraction

Lemma 9.3 (Euler's finite identity). *Let $c_1, \dots, c_N$ be nonzero and put $r_m = c_{m+1}/c_m$. Then*

$$\sum_{m=1}^N c_m = \frac{c_1}{1 + \frac{-r_1}{1 + r_1 + \frac{-r_2}{1 + r_2 + \frac{\ddots}{\ddots + \frac{-r_{N-1}}{1 + r_{N-1}}}}}}. \quad (9.6)$$

Proof. Write $R_m = c_m + \dots + c_N$. Starting from the bottom, one verifies by backward induction that the tail denominator beginning with $1 + r_m$ equals $r_m R_m / R_{m+1}$. Consequently the entire denominator in (9.6) is $1 - R_2/R_1 = c_1/R_1$, and the fraction equals R_1 . $\square$

Apply this identity to $c_n = (-n)^{n-1} x^n / n!$. A calculation gives

$$\frac{c_{m+1}}{c_m} = -\rho_m x, \quad \rho_m = \left(1 + \frac{1}{m}\right)^{m-1}.$$

Taking the limit of the finite identities therefore yields the following continued fraction wherever the Taylor series converges and no finite denominator vanishes:

$$W_0(x) = \frac{x}{1 + \frac{\rho_1 x}{1 - \rho_1 x + \frac{\rho_2 x}{1 - \rho_2 x + \frac{\rho_3 x}{1 - \rho_3 x + \ddots}}}}. \quad (9.7)$$

Interpreted as the limit of its finite truncations, this is simply a repackaging of the convergent Taylor series for $|x| < 1/e$.

Corcino, Corcino, and Mezo also derive a regular C-fraction beginning

$$W_0(x) = \frac{x}{1 + \frac{x}{1 + \frac{x/2}{1 + \frac{5x/6}{1 + \frac{17x/30}{1 + \ddots}}}}}, \quad (9.8)$$

with general coefficients expressed through Hankel determinants and computable by the quotient-difference algorithm [11]. Continued fractions can outperform a raw Taylor polynomial near a boundary of convergence, but their poles must be monitored. Again, a fraction derived from analyticity at zero represents W_0 , not W_{-1} .

9.5 Inverse-function integrals and moments

The substitution $x = we^w$ from Chapter 4 may itself be viewed as an integral representation. For any continuous F and any real branch segment on which the endpoints correspond to $w = a$ and $w = b$,

$$\int_{ae^a}^{be^b} F(W_k(x)) dx = \int_a^b F(w)e^w(1+w) dw. \quad (9.9)$$

The branch is encoded by the allowed w -interval: $[-1, \infty)$ for W_0 and $(-\infty, -1]$ for W_{-1} . Formula (9.9) is often a simpler representation than any integral whose integrand still contains W .

Chapter 10

Reliable numerical evaluation and practical approximations

No single truncated formula is uniformly excellent on an entire real branch. A reliable implementation therefore combines three ideas:

1. choose a local coordinate suited to the region;
2. choose the branch before iterating;
3. finish with a rapidly convergent correction whose iterates cannot jump to the other real branch.

The source archive contains several generations of this strategy, from the analytic approximations of Barry and coauthors [5], through the global principal-branch approximation of Iacono and Boyd [10], to the guaranteed high-precision analysis of Loczi [13] and the high-order Schroder expansions of Howard [14].

10.1 Newton and Halley applied directly

For

$$f_x(w) = we^w - x, \quad f'_x(w) = e^w(1 + w), \quad f''_x(w) = e^w(2 + w),$$

Newton's method is

$$w_{n+1} = w_n - \frac{w_n e^{w_n} - x}{e^{w_n}(1 + w_n)} = w_n - \frac{w_n - x e^{-w_n}}{1 + w_n}. \quad (10.1)$$

The second form avoids one exponential in the numerator, but it can still overflow or underflow when x and w_n are extreme.

Halley's method is

$$w_{n+1} = w_n - \frac{w_n e^{w_n} - x}{e^{w_n}(1 + w_n) - \frac{(w_n + 2)(w_n e^{w_n} - x)}{2(1 + w_n)}}. \quad (10.2)$$

At a simple root, Newton is locally quadratic and Halley is locally cubic.

Proposition 10.1. *Let f have three continuous derivatives near a simple zero r . For starts sufficiently near r , Newton's errors satisfy $e_{n+1} = (f''(r)/(2f'(r)))e_n^2 + \mathcal{O}(e_n^3)$, while Halley's errors are $\mathcal{O}(e_n^3)$.*

Proof. Write $y = r + e$ and expand

$$f(y) = ae + \frac{b}{2}e^2 + \frac{c}{6}e^3 + \mathcal{O}(e^4), \quad f'(y) = a + be + \frac{c}{2}e^2 + \mathcal{O}(e^3),$$

where $a = f'(r) \neq 0$. Division of the two series gives $f(y)/f'(y) = e - (b/2a)e^2 + \mathcal{O}(e^3)$, which proves the Newton formula. For Halley, substitute the same expansions into $y - 2ff'/(2(f')^2 - ff'')$; the coefficients of e and e^2 cancel, leaving an error of order e^3 . $\square$

The qualifier “simple” matters. At $x = -1/e$, the root $w = -1$ is double, $f'_x(-1) = 0$, and both displayed formulas have a singular denominator. Near that point the Puiseux coordinate is the right starting language.

10.2 A logarithmic Newton map

For real x and a real iterate b of the same sign, the equation $be^b = x$ is equivalent to

$$F_x(b) := b + \log |b| - \log |x| = 0. \quad (10.3)$$

Newton’s method for F_x is

$$T_x(b) = \frac{b}{1+b} \left(1 + \log \frac{x}{b} \right). \quad (10.4)$$

The ratio x/b is positive, so the logarithm is real on both real branches. This iteration was emphasized by Iacono and Boyd and analyzed globally with explicit error bounds by Loczi [10, 13]. It needs no exponential and is well scaled for $W_0(x)$ at huge positive x and for $W_{-1}(x)$ at tiny negative x .

The following theorem gives a self-contained branch-safe convergence result.

Theorem 10.2 (three monotone invariant intervals). *Let w be the indicated real solution of $we^w = x$.*

- (a) *If $w > 0$ and $0 < b_0 < w$, then $0 < b_0 < b_1 < \dots < w$, and $b_n \rightarrow w$.*
- (b) *If $-1 < w < 0$ and $w < b_0 < 0$, then $w < \dots < b_2 < b_1 < b_0 < 0$, and $b_n \rightarrow w$.*
- (c) *If $w < -1$ and $b_0 < w$, then $b_0 < b_1 < b_2 < \dots < w$, and $b_n \rightarrow w$.*

Here $b_{n+1} = T_x(b_n)$.

Proof. For any admissible b , put $r = b/w > 0$. Since $x = we^w$, direct simplification gives the two exact identities

$$T_x(b) - w = \frac{w(r - 1 - r \log r)}{1 + b}, \quad (10.5)$$

$$T_x(b) - b = \frac{b}{1 + b} \left(\log \frac{w}{b} + w - b \right). \quad (10.6)$$

The function $h(r) = r - 1 - r \log r$ has derivative $-\log r$ and unique maximum $h(1) = 0$. Thus $h(r) < 0$ when $r \neq 1$.

In case (a), $0 < r < 1$. Equation (10.5) gives $T_x(b) < w$, while both terms in the parentheses of (10.6) are positive, so $T_x(b) > b$.

In case (b), again $0 < r < 1$. Now $w < 0$ and $1 + b > 0$, so (10.5) gives $T_x(b) > w$. Moreover

$$\log \frac{w}{b} + w - b = -\log r + w(1 - r) > (1 - r) - (1 - r) = 0,$$

because $-\log r > 1 - r$ and $w > -1$. Since $b/(1 + b) < 0$, (10.6) gives $T_x(b) < b$.

In case (c), $r > 1$, $w < 0$, and $1 + b < 0$, so (10.5) gives $T_x(b) < w$. Also

$$\log \frac{w}{b} + w - b = -\log r + (-w)(r - 1) > -\log r + r - 1 > 0.$$

Here $b/(1+b) > 0$, hence $T_x(b) > b$.

Each sequence is monotone and bounded, so it has a limit ℓ . Passing to the limit in (10.4) gives $\log(x/\ell) = \ell$, hence $\ell e^\ell = x$. The invariant interval contains only the specified real root, so $\ell = w$. $\square$

Corollary 10.3 (quadratic error constant). *For $b = w + e$ near a real root $w \neq -1, 0$,*

$$T_x(b) - w = -\frac{e^2}{2w(1+w)} + \mathcal{O}(e^3). \quad (10.7)$$

Thus the convergence in Theorem 10.2 is quadratic once the iterates are near the root.

Proof. Set $r = 1 + e/w$ in (10.5) and use

$$r - 1 - r \log r = -\frac{1}{2}(r - 1)^2 + \frac{1}{6}(r - 1)^3 + \mathcal{O}((r - 1)^4).$$

Expanding $1/(1 + w + e)$ gives (10.7). $\square$

10.3 Explicit global starting values

Algorithm 10.4 (branch-safe real starts). Handle $x = -1/e$, $x = 0$, and $x = e$ exactly. Otherwise use

$$b_0 = x, \quad -1/e < x < 0, \quad W_0, \quad (10.8)$$

$$b_0 = x/e, \quad 0 < x < e, \quad W_0, \quad (10.9)$$

$$b_0 = \log x - \log \log x, \quad x > e, \quad W_0, \quad (10.10)$$

$$b_0 = -1 - \sqrt{2s} - s, \quad s = -1 - \log(-x), \quad -1/e < x < 0, \quad W_{-1}. \quad (10.11)$$

Then iterate (10.4).

Theorem 10.5. *Every start in Theorem 10.4 satisfies the corresponding hypothesis of Theorem 10.2; consequently it converges monotonically to the requested branch value.*

Proof. For (10.8), if $w = W_0(x) \in (-1, 0)$ then $x = we^w > w$, so $w < x < 0$. For (10.9), W_0 is strictly concave on $(0, \infty)$, and the chord joining $(0, 0)$ and $(e, 1)$ is x/e ; hence $0 < x/e < W_0(x)$ for $0 < x < e$.

For (10.10), put $A = \log x > 1$ and $b_0 = A - \log A > 1$. Since $b_0 e^{b_0} = x b_0 / A < x$ and $w \mapsto we^w$ is increasing for $w > -1$, we have $b_0 < W_0(x)$.

Finally (10.11) is precisely the strict lower bound in (8.8). The three cases now follow from Theorem 10.2. $\square$

The starts are deliberately simple rather than minimax-optimal. Loczi gives piecewise starts with explicit uniform errors that square at every step; Howard constructs generic Schroder series from an initial approximating function and reports extremely high accuracy at high order [13, 14].

10.4 A posteriori error certification

An iterate should be stopped because its error is bounded, not merely because successive floating-point numbers happen to be equal.

Proposition 10.6. *Let I be an interval containing a unique root w of F_x in (10.3). If $|F'_x(t)| \geq m > 0$ throughout I , then every $b \in I$ satisfies*

$$|b - w| \leq \frac{|F_x(b)|}{m}, \quad F'_x(t) = 1 + \frac{1}{t}. \quad (10.12)$$

Proof. The mean-value theorem gives $F_x(b) - F_x(w) = F'_x(\xi)(b - w)$ for some ξ between b and w . Take absolute values and use $F_x(w) = 0$. $\square$

For $W_0(x) > 0$, one may take the crude universal value $m = 1$. On either negative branch, $|1 + 1/t|$ tends to zero as $t \rightarrow -1$, so no residual bound in the original w coordinate can remain uniform at the branch point. There, the p coordinate of Chapter 6 supplies the appropriate conditioning.

A robust implementation can maintain a one-sided monotone bound from Theorem 10.2, compute a second bound from an analytic inequality, and then stop when the bracket width is below the requested tolerance. For ordinary machine precision, the combination of a region-aware start and two or three Halley or logarithmic-Newton corrections is generally more than sufficient.

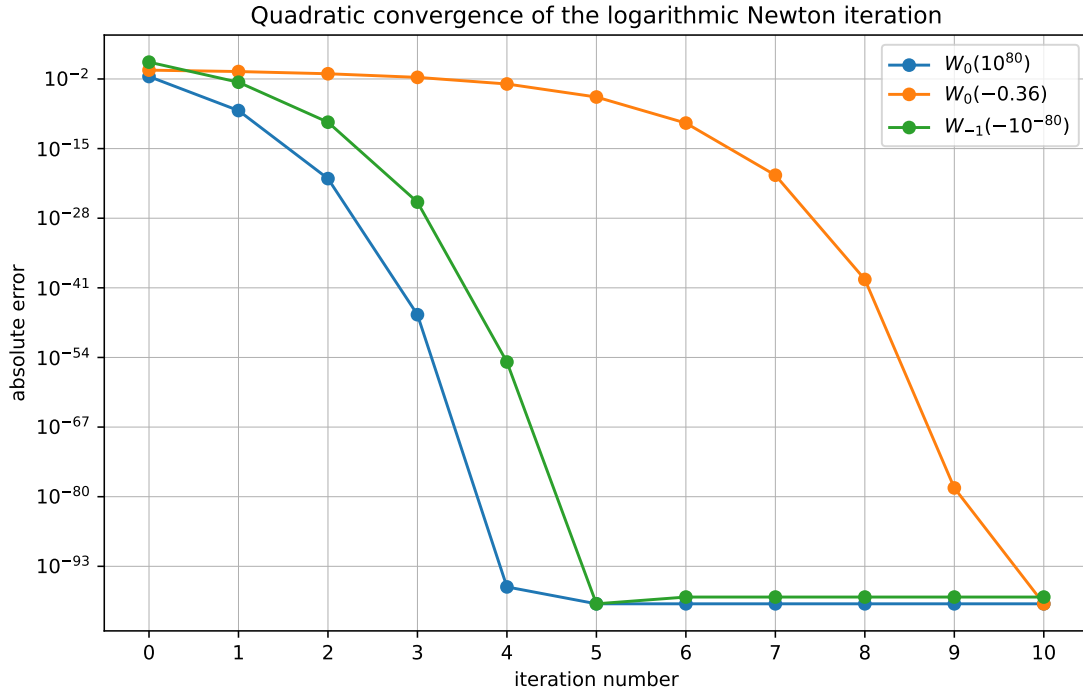


Figure 10.1: Absolute error under the logarithmic Newton iteration for a huge positive argument, a point close to the branch point on W_0 , and a tiny negative argument on W_{-1} . The slopes steepen in accordance with quadratic convergence until the working precision is reached.

10.5 Choosing an approximation by region

Region	Natural coordinate	Good first approximation	Main caution
$W_0, x \ll 1$	x	Taylor series (5.4)	radius is exactly $1/e$
Either branch near $-1/e$	$p = \sqrt{2(1+ex)}$	paired Puiseux series (6.8), (6.9)	subtraction in $1+ex$
$W_0, x \gg 1$	$(\log x, \log \log x)$	(7.5)	asymptotic, not a small- x series
$W_{-1}, x \uparrow 0$	$\eta = \log(1/(-x))$	(7.12)	preserve the lower branch and the sign
Middle of a branch	w itself	monotone start plus (10.4)	never cross $w = -1$

A practical composite evaluator is therefore:

1. use a Puiseux truncation when $1+ex$ is small;
2. otherwise use a Taylor or logarithmic truncation when its natural small parameter is small;
3. otherwise use Theorem 10.4;
4. apply logarithmic Newton or Halley until a residual or bracket test passes.

The accompanying program implements and tests these ingredients. It uses high-precision reference values only to produce the figures; the formulas being tested are coded explicitly.

10.6 Published global approximations in the archive

It is useful to distinguish an *analytic approximation*, intended to be inserted into further algebra, from a *numerical seed*, intended only to start an iteration.

Barry, Parlange, Li, Prommer, Cunningham, and Stagnitti construct separate closed forms for the three real pieces W_{-1} , $W_0|_{[-1/e,0]}$, and $W_0|_{[0,\infty)}$, arranging exact endpoint behavior and small reported maximum relative errors [5]. This separation is mathematically natural: no expression analytic at zero can simultaneously represent the unbounded branch W_{-1} .

Iacono and Boyd give a simple global approximation to W_0 with reported maximum relative error below 5×10^{-3} , then reach machine precision with three steps of the logarithmic Newton map (10.4); they also derive tight large- x bounds and discuss Chebyshev approximations [10]. Chebyshev polynomials work best after a change of variable that resolves both the square-root singularity at $-1/e$ and the logarithmic growth at infinity.

Howard's Schroder-type construction starts with an arbitrary approximating function lying inside proved upper and lower envelopes and generates an inverse-function series of arbitrary order for either real branch [14]. It provides a systematic bridge between a compact closed-form seed and very high precision. The conceptual lesson is broader than any one formula: a good coordinate and a certified basin are more important than a long polynomial in the original variable.

10.7 Floating-point hazards

1. Near $-1/e$, the expression $1+ex$ loses relative accuracy by cancellation. If the input is naturally $x = -1/e + h$, compute p from h ; otherwise use a fused multiply-add when available.
2. Near $x = 0^-$ on W_{-1} , forming x may underflow even though $\log(-x)$ is representable. An interface accepting $\log(-x)$ directly is more robust.

3. In (10.4), evaluate $\log(x/b)$ as $\log|x| - \log|b|$ when the ratio would overflow or underflow.
4. Relative error in $W_0(x)$ is ill-conditioned at $x = 0$, where the true value vanishes. Use absolute error there.
5. A tiny residual $we^w - x$ does not by itself guarantee a tiny error near the double root at $-1/e$; account for the small derivative or use the Puiseux coordinate.

Chapter 11

Complex branches and analytic structure

The two real graphs are only boundary traces of a countable family of complex inverse branches. This chapter gives the minimum complex analysis needed to understand where those branches come from and why the real singularities have the forms already derived.

11.1 Critical and asymptotic values of we^w

Let $f(w) = we^w$. It is entire and

$$f'(w) = e^w(1 + w).$$

Thus its only finite critical point is $w = -1$, and the corresponding critical value is $-1/e$. The local expansion

$$f(-1 + u) = -\frac{1}{e} + \frac{u^2}{2e} + \frac{u^3}{3e} + \cdots$$

starts with u^2 ; locally, inversion must therefore introduce a square root. This recovers the Puiseux behavior of Chapter 6 without any real-variable graph.

There is also an inverse singularity over $z = 0$. Although $f(0) = 0$ gives the regular principal value $W_0(0) = 0$, one has $we^w \rightarrow 0$ along paths with $\operatorname{Re} w \rightarrow -\infty$. The nonprincipal inverse branches therefore run off to infinity as $z \rightarrow 0$, producing logarithmic branch points.

11.2 Standard branch domains

The analytic inverse-function theorem gives a local holomorphic inverse at every point where $w \neq -1$. Global single-valued branches require cuts. Under the standard convention used by Corless et al. and the DLMF [3, 4]:

- W_0 is analytic on $\mathbb{C} \setminus (-\infty, -1/e]$ and is real on $(-1/e, \infty)$;
- the nonprincipal branches W_k , $k \neq 0$, are represented on cut planes with a cut ending at 0; they have logarithmic behavior there;
- the sheets adjacent to W_0 meet it at the square-root point $-1/e$. The real function W_{-1} is the lower real boundary value on $(-1/e, 0)$.

Different software systems may place boundary values on different lips of a cut. An index k is therefore meaningful only together with a stated logarithm convention.

11.3 Why there are infinitely many branches

Taking a logarithm of $we^w = z$ gives, formally,

$$w + \operatorname{Log} w = \operatorname{Log} z + 2\pi in, \quad n \in \mathbb{Z}. \quad (11.1)$$

For large $|n|$, the right side has imaginary part approximately $2\pi n$, and a solution exists near it. The asymptotic expansion (7.18) makes this precise. In particular,

$$W_{k+1}(z) - W_k(z) = 2\pi i + \mathcal{O}\left(\frac{1}{k}\right) \quad (|k| \rightarrow \infty), \quad (11.2)$$

for fixed nonzero z away from cuts.

A circuit around $z = 0$ changes $\operatorname{Log} z$ by $2\pi i$. Equation (11.1) then moves the analytic continuation to a neighboring sheet. This phenomenon is called *monodromy*; the branch index records how many logarithmic turns have been made.

11.4 Local behavior at zero on nonprincipal branches

Let

$$L_{1,k} = \operatorname{Log} z + 2\pi ik, \quad L_{2,k} = \operatorname{Log} L_{1,k}.$$

Then, as $z \rightarrow 0$ along a fixed sector on a nonprincipal sheet,

$$W_k(z) = L_{1,k} - L_{2,k} + \frac{L_{2,k}}{L_{1,k}} + \mathcal{O}\left(\frac{L_{2,k}^2}{L_{1,k}^2}\right). \quad (11.3)$$

This is the same logarithmic inversion mechanism as at large z ; what matters is that $|L_{1,k}|$ is large. The real lower-branch formula (7.14) is the cut-boundary version of this statement, rewritten with entirely real logarithms.

11.5 Conjugation and real coefficients

Because $f(\bar{w}) = \overline{f(w)}$, complex solutions occur in conjugate pairs. On any conjugation-symmetric region avoiding the cuts, branch labels can be chosen so that

$$\overline{W_k(z)} = W_{-k}(\bar{z}). \quad (11.4)$$

For the principal branch this reduces to $\overline{W_0(z)} = W_0(\bar{z})$. On a cut one must specify upper and lower boundary values; conjugation interchanges the two lips.

11.6 A transcendence consequence

Theorem 11.1. *If $z \neq 0$ is algebraic, then every value $W_k(z)$ is transcendental.*

Proof. Suppose $w = W_k(z)$ were algebraic. It is nonzero because $z \neq 0$. The Lindemann–Weierstrass theorem says that e^w is transcendental for nonzero algebraic w . But $we^w = z$ gives $e^w = z/w$, which would be algebraic. This contradiction proves the claim. $\square$

Thus even the omega constant $W_0(1)$ is transcendental. The theorem does not say that values at transcendental arguments must be transcendental; special cancellations can occur, as the next chapter shows.

Chapter 12

Identities, branch exchange, and special values

Many attractive Lambert identities are really statements of the form “these two expressions have the same product with their exponential.” The last step must always specify which inverse branch returns the proposed value.

12.1 Values generated from a chosen output

For every complex a ,

$$W_k(ae^a) = a \quad (12.1)$$

whenever the k th branch is the inverse sheet containing a . On the real axis this specializes to

$$W_0(ae^a) = a \quad (a \geq -1), \quad (12.2)$$

$$W_{-1}(ae^a) = a \quad (a \leq -1). \quad (12.3)$$

This is the safest way to manufacture exact values. Examples are

$$\begin{aligned} W_0(0) &= 0, & W_0(e) &= 1, & W_0(-1/e) &= W_{-1}(-1/e) = -1, \\ W_{-1}(-2e^{-2}) &= -2, & W_{-1}(-3e^{-3}) &= -3. \end{aligned}$$

The principal value $\Omega = W_0(1)$ is the unique positive solution of $\Omega e^\Omega = 1$, equivalently $\Omega = e^{-\Omega}$.

A complex example is

$$\left(\frac{\pi i}{2}\right) \exp\left(\frac{\pi i}{2}\right) = -\frac{\pi}{2}. \quad (12.4)$$

Thus $\pi i/2$ and $-\pi i/2$ are conjugate Lambert values of $-\pi/2$ on the two sides of the principal cut.

12.2 The real branch-exchange involution

The function $q \mapsto qe^{-q}$ increases on $(0, 1]$ and decreases on $[1, \infty)$. Therefore every $0 < y < 1/e$ has two positive preimages, one on each side of 1.

Definition 12.1. For $0 < t \leq 1$, define

$$\sigma(t) = -W_{-1}(-te^{-t}). \quad (12.5)$$

Then $\sigma(t) \geq 1$ and $te^{-t} = \sigma(t)e^{-\sigma(t)}$.

Conversely, for $s \geq 1$,

$$-W_0(-se^{-s}) \in (0, 1] \quad (12.6)$$

returns the partner on the other side. These operations are inverses. They make explicit that W_0 and W_{-1} are not unrelated functions: on the negative real interval they encode the two inverse sides of the same unimodal map.

12.3 An addition identity with a branch condition

Proposition 12.2. *Let $a = W_j(x)$ and $b = W_k(y)$ be nonzero. Then*

$$(a + b)e^{a+b} = xy \left(\frac{1}{a} + \frac{1}{b} \right). \quad (12.7)$$

Consequently,

$$a + b = W_\ell \left(xy \left(\frac{1}{a} + \frac{1}{b} \right) \right) \quad (12.8)$$

for whichever branch ℓ contains $a + b$.

Proof. Since $e^a = x/a$ and $e^b = y/b$,

$$(a + b)e^{a+b} = (a + b) \frac{x}{a} \frac{y}{b} = xy \left(\frac{1}{a} + \frac{1}{b} \right).$$

Applying the inverse relation gives (12.8), with the branch qualification forced by multivaluedness. $\square$

The identity is algebraically exact but usually not computationally useful, because its right side already contains a and b . Its value is structural: it shows how ordinary addition in the w -plane is transported through the map $w \mapsto we^w$.

12.4 Scaling identities

If $w = W_k(x)$ and n is a positive integer, then

$$(nw)e^{nw} = \frac{nx^n}{w^{n-1}}. \quad (12.9)$$

Hence

$$nW_k(x) = W_\ell \left(\frac{nx^n}{W_k(x)^{n-1}} \right) \quad (12.10)$$

for the branch containing $nW_k(x)$. More generally, for any complex α ,

$$(\alpha w)e^{\alpha w} = \alpha x^\alpha w^{1-\alpha},$$

provided powers and logarithms are chosen consistently.

12.5 Fixed points of W

If a branch value satisfies $W_k(z) = z$, then $ze^z = z$. Therefore either $z = 0$ or $e^z = 1$, giving

$$z = 2\pi in, \quad n \in \mathbb{Z}. \quad (12.11)$$

Every candidate indeed satisfies the inverse relation, but the branch index that returns it depends on the chosen cuts. On the real principal branch the only fixed point is 0.

12.6 A branch-aware logarithm identity

From $W_k(z)e^{W_k(z)} = z$ one often writes

$$W_k(z) = \operatorname{Log} z - \operatorname{Log} W_k(z) + 2\pi im. \quad (12.12)$$

The integer m is not optional: it records the discrepancy between the selected logarithms. On $W_0(x)$ for $x > 0$, both values are positive and $m = 0$. On the two negative real branches, taking real logarithms of absolute values gives the unambiguous equations

$$W_0(x) + \log(-W_0(x)) = \log(-x), \quad (12.13)$$

$$W_{-1}(x) + \log(-W_{-1}(x)) = \log(-x), \quad (12.14)$$

with the two roots separated by -1 .

Chapter 13

Applications derived step by step

The point of an application is not merely to display the symbol W . One should see exactly how the equation is transformed, how many real solutions exist, and why a particular branch is admissible.

13.1 A branch-selection checklist

When a model reduces to $ue^u = z$:

1. determine the real range required of u before applying W ;
2. if $-1/e < z < 0$, write both candidates $W_0(z)$ and $W_{-1}(z)$;
3. translate each candidate back to the original variable;
4. reject a candidate only for a mathematical or modeling reason—wrong sign, wrong interval, boundary solution, or violation of an earlier division;
5. check the result in the original equation.

Several examples below contain a tempting but physically irrelevant second root. The branch analysis explains it rather than hiding it.

13.2 Characteristic exponents of a delay differential equation

Consider

$$y'(t) = a y(t - \tau), \quad \tau > 0. \quad (13.1)$$

Try a mode $y(t) = e^{\lambda t}$. Substitution gives

$$\lambda e^{\lambda t} = a e^{\lambda(t-\tau)}, \quad \lambda = a e^{-\lambda \tau}.$$

Therefore

$$(\lambda \tau) e^{\lambda \tau} = a \tau, \quad \boxed{\lambda_k = \frac{1}{\tau} W_k(a \tau)}, \quad k \in \mathbb{Z}. \quad (13.2)$$

The infinitely many complex branches give the infinitely many characteristic exponents of the delay equation, one reason Lambert W became important in this subject [3].

If $-1/e < a\tau < 0$, two exponents are real: $W_0(a\tau)/\tau \in (-1/\tau, 0)$ and $W_{-1}(a\tau)/\tau < -1/\tau$. If $a\tau < -1/e$, no characteristic exponent is real, although complex conjugate pairs remain. Determining stability of a general delay equation requires the full spectrum, not only the principal root.

13.3 A diode with series resistance

The ideal diode law with ideality factor n_d , thermal voltage V_T , reverse saturation current I_s , and series resistance R_s is

$$I = I_s \left(\exp \frac{V_D}{n_d V_T} - 1 \right), \quad V = V_D + R_s I. \quad (13.3)$$

Eliminate $V_D = V - R_s I$ and put $J = I + I_s$. Then

$$\begin{aligned} J &= I_s \exp \left(\frac{V - R_s(J - I_s)}{n_d V_T} \right) \\ &= I_s \exp \left(\frac{V + R_s I_s}{n_d V_T} \right) \exp \left(-\frac{R_s J}{n_d V_T} \right). \end{aligned}$$

Multiplying by the last exponential's inverse gives

$$\frac{R_s J}{n_d V_T} \exp \left(\frac{R_s J}{n_d V_T} \right) = \frac{R_s I_s}{n_d V_T} \exp \left(\frac{V + R_s I_s}{n_d V_T} \right).$$

Hence

$$\boxed{I = \frac{n_d V_T}{R_s} W_0 \left(\frac{R_s I_s}{n_d V_T} \exp \frac{V + R_s I_s}{n_d V_T} \right) - I_s.} \quad (13.4)$$

The argument is positive, so only W_0 is real. As $R_s \rightarrow 0$, the argument tends to zero and $W_0(z) \sim z$, recovering the ordinary ideal-diode law. Banwell used this formula and its transistor analogues to turn several implicit circuit relations into explicit ones [6].

13.4 Integrated Michaelis–Menten kinetics

Let a substrate concentration $S(t)$ satisfy

$$\frac{dS}{dt} = -\frac{V_{\max} S}{K_M + S}, \quad S(0) = S_0 > 0. \quad (13.5)$$

Separating variables and integrating gives

$$S + K_M \log S = S_0 + K_M \log S_0 - V_{\max} t. \quad (13.6)$$

Set $y = S/K_M$. After dividing by K_M and exponentiating,

$$y e^y = \frac{S_0}{K_M} \exp \left(\frac{S_0 - V_{\max} t}{K_M} \right).$$

Therefore

$$\boxed{S(t) = K_M W_0 \left(\frac{S_0}{K_M} \exp \frac{S_0 - V_{\max} t}{K_M} \right).} \quad (13.7)$$

Again the argument is positive, and positivity of concentration selects W_0 . The formula permits direct differentiation with respect to kinetic parameters and avoids repeatedly solving the implicit integrated equation. Golicnik surveys this and several modified biochemical rate laws [7].

13.5 The final-size equation of an epidemic

In the simplest normalized SIR model, the limiting susceptible fraction in the infinitesimal-seed idealization obeys

$$s_\infty = \exp(-R_0(1 - s_\infty)). \quad (13.8)$$

Rearrange it as

$$(-R_0 s_\infty) \exp(-R_0 s_\infty) = -R_0 e^{-R_0}.$$

Thus

$$s_\infty = -\frac{1}{R_0} W_k(-R_0 e^{-R_0}). \quad (13.9)$$

There is always a formal root $s_\infty = 1$, because one Lambert value is $-R_0$.

If $0 < R_0 < 1$, then $-R_0 \in (-1, 0)$ and $W_0(-R_0 e^{-R_0}) = -R_0$; the principal branch gives $s_\infty = 1$, while the lower branch gives a value exceeding 1 and is inadmissible. If $R_0 > 1$, then $-R_0 < -1$ and it is W_{-1} that returns the trivial root $-R_0$. The nontrivial epidemic root is

$$s_\infty = -\frac{1}{R_0} W_0(-R_0 e^{-R_0}) \in (0, 1/R_0). \quad (13.10)$$

At $R_0 = 1$, the two branches coalesce at $-1/e$. The epidemic threshold is therefore literally the Lambert branch point. Models with nonzero initial infected fraction modify the constant in (13.8), but the same branch-aware algebra applies; ecological and epidemiological examples are discussed by Lehtonen [9].

13.6 Optimal residence time in a resource patch

Suppose the cumulative gain in a patch after time t is

$$G(t) = A(1 - e^{-bt}), \quad A, b > 0,$$

and travel to the next patch takes time $\tau > 0$. The long-run gain rate is $G(t)/(t + \tau)$. An interior optimum satisfies

$$G'(t)(t + \tau) = G(t).$$

After cancelling A and multiplying by e^{bt} ,

$$e^{bt} = 1 + b(t + \tau). \quad (13.11)$$

Put $y = 1 + b(t + \tau) > 1$. Then $bt = y - 1 - b\tau$, so

$$ye^{-y} = e^{-1-b\tau}, \quad -y = W_k(-e^{-1-b\tau}).$$

Because $y > 1$, one must choose the lower branch:

$$t_* = \frac{-W_{-1}(-e^{-1-b\tau}) - 1 - b\tau}{b}. \quad (13.12)$$

The principal branch would give $0 < y < 1$ and hence a negative residence time. This is a clean biological example in which omitting W_{-1} loses the only admissible solution [9].

13.7 Wien's displacement constant

Up to constants independent of wavelength, Planck's spectral radiance is

$$B_\lambda(\lambda, T) \propto \frac{\lambda^{-5}}{\exp\left(\frac{hc}{\lambda k_B T}\right) - 1}.$$

Set $x = hc/(\lambda k_B T) > 0$. Maximizing in λ is equivalent to maximizing $x^5/(e^x - 1)$. Its logarithmic derivative vanishes when

$$\frac{5}{x} - \frac{e^x}{e^x - 1} = 0, \quad (5 - x)e^x = 5.$$

With $u = x - 5$,

$$ue^u = -5e^{-5}, \quad x = 5 + W_k(-5e^{-5}).$$

The lower branch gives $W_{-1}(-5e^{-5}) = -5$ and hence the boundary root $x = 0$. The positive interior maximum comes from the principal branch:

$$\boxed{x_* = 5 + W_0(-5e^{-5}) \approx 4.965114231744276.} \quad (13.13)$$

Thus $\lambda_{\max} T = hc/(k_B x_*)$. The two branches neatly separate the spurious boundary stationary equation from the physical maximum.

13.8 Time for a falling body with linear drag

Take downward distance as positive. A body released from rest with drag coefficient per unit mass $\gamma > 0$ and terminal speed $v_\infty = g/\gamma$ travels

$$s(t) = v_\infty \left(t - \frac{1 - e^{-\gamma t}}{\gamma} \right). \quad (13.14)$$

Given $s \geq 0$, set

$$y = \gamma t, \quad A = 1 + \frac{\gamma s}{v_\infty} \geq 1.$$

Equation (13.14) becomes $y + e^{-y} = A$. Put $v = A - y = e^{-y}$; then $(-v)e^{-v} = -e^{-A}$ and

$$\boxed{t = \frac{A + W_0(-e^{-A})}{\gamma}.} \quad (13.15)$$

For $A > 1$, the principal value lies in $(-1, 0)$ and gives $t > 0$. The W_{-1} value is less than $-A$ and gives a negative-time root of the algebraically extended equation. At $A = 1$ the branches meet and $t = 0$.

13.9 Inverting the Schwarzschild tortoise coordinate

For $r > 2M$, the Schwarzschild tortoise coordinate is

$$r_* = r + 2M \log \left(\frac{r}{2M} - 1 \right). \quad (13.16)$$

Set $y = r/(2M) - 1 > 0$. Then

$$y + \log y = \frac{r_*}{2M} - 1, \quad ye^y = \exp\left(\frac{r_*}{2M} - 1\right).$$

Consequently

$$\boxed{r = 2M \left[1 + W_0\left(\exp\left(\frac{r_*}{2M} - 1\right)\right) \right]}. \quad (13.17)$$

The argument is positive, so the exterior coordinate uses only W_0 . The Wright omega form can evaluate this expression without overflowing when $r_*/(2M)$ is large.

13.10 Prime counting and the n th prime

Let p_n be the n th prime and $\pi(x)$ the number of primes not exceeding x . A classical inequality states $p_n > n \log n$ for $n \geq 1$. If $x \geq 2$ and $n = \pi(x)$, then $p_n \leq x$, so

$$n \log n < x.$$

The inverse of $u \mapsto u \log u$ on $[1, \infty)$ is $u = e^{W_0(x)} = x/W_0(x)$. Hence

$$\boxed{\pi(x) < \frac{x}{W_0(x)} = e^{W_0(x)} \quad (x \geq 2)}. \quad (13.18)$$

Since $W_0(x) \sim \log x$, the prime number theorem $\pi(x) \sim x/\log x$ is equivalently $\pi(x) \sim x/W_0(x)$. Visser develops this viewpoint and related bounds [12].

The lower branch appears when the prime-number theorem is inverted. Solving

$$n = \frac{x}{\log x}, \quad x > e,$$

put $y = \log x > 1$. Then $e^y/y = n$, so

$$(-y)e^{-y} = -\frac{1}{n}.$$

The requirement $y > 1$ selects W_{-1} :

$$\boxed{x = -nW_{-1}(-1/n)}. \quad (13.19)$$

Thus the first-order inversion of the prime number theorem suggests

$$p_n \approx -nW_{-1}(-1/n). \quad (13.20)$$

Inserting the lower-branch asymptotic expansion gives $n(\log n + \log \log n + \dots)$ automatically. Choosing W_0 instead would give $0 < y < 1$ and the wrong solution $x < e$.

13.11 Rooted labeled trees revisited

Chapter 5 showed that

$$-W_0(-z) = \sum_{n \geq 1} \frac{n^{n-1}}{n!} z^n.$$

The coefficient n^{n-1} counts rooted labeled trees on n vertices. This is more than a numerical coincidence: a rooted labeled tree is a root attached to an unordered set of rooted labeled trees, and the exponential formula for labeled structures translates that recursive description into $T = ze^T$. Lambert inversion therefore packages the entire recursive combinatorial class in one special function.

13.12 What the examples have in common

Every derivation above used one of three normal forms:

$$(ax + b)e^{cx+d} = r, \quad x + \log x = c, \quad x - \log x = c.$$

The first is solved directly by W ; the second is naturally expressed by W_0 or Wright omega; the third is the real lower-branch inversion near zero. Recognizing these forms is usually more important than memorizing any application-specific formula.

Chapter 14

Generalizations and neighboring inverse functions

Lambert W is the simplest non-elementary inverse created by mixing a polynomial factor with an exponential. More complicated implicit equations lead to related functions rather than to new identities for ordinary W .

14.1 The r -Lambert function

The r -Lambert function W_r is defined by

$$we^w + rw = z. \quad (14.1)$$

At $r = 0$ it reduces to ordinary W . Implicit differentiation gives

$$W'_r(z) = \frac{1}{e^{W_r(z)}(1 + W_r(z)) + r}, \quad (14.2)$$

whenever the denominator is nonzero. Its branch points are images of the critical points satisfying

$$e^w(1 + w) + r = 0.$$

Unlike ordinary W , these critical points themselves generally require a Lambert function to describe. The same workflow remains valid: locate critical points, divide into monotonic intervals, and assign inverse branches.

14.2 Products of shifted linear factors

Equations such as

$$e^x \frac{(x - t_1) \cdots (x - t_p)}{(x - s_1) \cdots (x - s_q)} = a \quad (14.3)$$

lead to generalized Lambert functions. Ordinary W is the case with one unshifted numerator factor and no denominator factors. Some special choices of the shifts reduce to ordinary W after an affine substitution, but in general no finite combination of ordinary branches isolates x .

The lesson is diagnostic. If all appearances of the unknown can be gathered into a single affine factor times the exponential of an affine factor, [Theorem 2.1](#) applies. If several independent affine factors remain, a genuine generalization is usually required.

14.3 Wright omega as a numerically normalized inverse

The equation

$$y + \log y = z$$

is mathematically equivalent to $y = W(e^z)$, but the omega form is often a better function in its own right. It removes an unnecessary exponential, makes the dominant large- z behavior $y \sim z$ explicit, and handles complex logarithm unwinding in a systematic way. This illustrates a general principle in special-function design: equivalent functions can have very different numerical conditioning.

14.4 Inverse functions and root-finding series

Schroder's theory constructs high-order inverse approximations from the root-finding problem $f(w) - z = 0$. Newton, Halley, and higher Householder maps are finite-order members of the same broad family. Howard's recent treatment in the archive applies this viewpoint to both real Lambert branches [14]. The branch-independent part is algebraic differentiation; the branch-dependent part is the certified initial approximation and the avoidance of critical points.

14.5 Questions for further study

Several directions remain approachable after this article:

- derive minimax rational approximations after mapping each unbounded branch to a finite interval;
- obtain interval-arithmetic implementations whose enclosures remain sharp at the square-root point;
- study how continued-fraction poles interlace with the branch cut;
- develop real-variable analogues of complex branch monodromy for parameter-dependent models where two physical roots merge;
- compare ordinary W , Wright omega, and r -Lambert formulations for stiff delay equations.

Chapter 15

Exercises with complete solutions

The exercises are ordered roughly by chapter. Each solution is included so that the chapter can be used for self-study.

15.1 Problems

Exercise 15.1. For each real c , determine the number of real solutions of $xe^x = c$ and identify the relevant Lambert branches.

Exercise 15.2. Solve $(2x - 1)e^{3x} = 4$ in Lambert form and determine how many real solutions it has.

Exercise 15.3. Solve $x^x = 0.9$ for positive x . Explain why there are two solutions and write both in terms of W .

Exercise 15.4. Compute $W_0^{(n)}(0)$ for $1 \leq n \leq 6$.

Exercise 15.5. Evaluate

$$\int_{-1/e}^0 W_0(x) dx, \quad \int_{-1/e}^0 W_{-1}(x) dx.$$

Why does the second improper integral converge even though its integrand tends to $-\infty$?

Exercise 15.6. Starting from $w = -1 + u$, show by coefficient matching that

$$W_0(x) = -1 + p - \frac{1}{3}p^2 + \mathcal{O}(p^3), \quad p = \sqrt{2(1 + ex)}.$$

Obtain the corresponding formula for W_{-1} .

Exercise 15.7. Starting from $w + \log w = \log x$, derive

$$W_0(x) = L_1 - L_2 + \frac{L_2}{L_1} + \mathcal{O}\left(\frac{L_2^2}{L_1^2}\right).$$

Exercise 15.8. Prove directly that

$$W_{-1}(x) \leq \log(-x) - \log(-\log(-x)) \quad (-1/e \leq x < 0).$$

Exercise 15.9. Let $x > e$ and $b_0 = \log x - \log \log x$. Prove without invoking [Theorem 10.5](#) that one logarithmic Newton step satisfies $b_0 < T_x(b_0) < W_0(x)$.

Exercise 15.10. For $y'(t) = -y(t-1)/4$, find the two real characteristic exponents and state which branch gives each one.

Exercise 15.11. For the patch model of (13.12), take $b = 1$ and $\tau = 1$. Write the exact optimal residence time and show it is positive.

Exercise 15.12. Solve $x/\log x = n$ for $x > e$ and show that the principal branch gives the wrong inverse.

Exercise 15.13. Use Lagrange inversion to prove $[z^n]T(z)^r = rn^{n-r-1}/(n-r)!$ for integers $n \geq r \geq 1$, where $T = ze^T$.

15.2 Solutions

Solution to Theorem 15.1. The function $f(x) = xe^x$ decreases from 0^- to $-1/e$ on $(-\infty, -1]$ and increases from $-1/e$ to ∞ on $[-1, \infty)$. Hence there are no real solutions for $c < -1/e$, one solution $x = -1$ for $c = -1/e$, two solutions $W_{-1}(c) < -1 < W_0(c) < 0$ for $-1/e < c < 0$, one solution $x = 0$ for $c = 0$, and one solution $W_0(c) > 0$ for $c > 0$.

Solution to Theorem 15.2. Apply Theorem 2.1 with $a = 2$, $b = -1$, $c = 3$, $d = 0$, and $r = 4$:

$$x = \frac{1}{3}W_k\left(6e^{3/2}\right) + \frac{1}{2}.$$

The argument is positive, so only W_0 is real and there is exactly one real solution.

Solution to Theorem 15.3. Here $\log 0.9 \in (-1/e, 0)$ because $0.9 > e^{-1/e}$. Therefore

$$x_1 = \exp(W_0(\log 0.9)), \quad x_2 = \exp(W_{-1}(\log 0.9)).$$

They lie on opposite sides of e^{-1} , the minimum point of x^x .

Solution to Theorem 15.4. By (5.5),

$$W'_0(0) = 1, \quad W''_0(0) = -2, \quad W_0^{(3)}(0) = 9, \quad W_0^{(4)}(0) = -64, \quad W_0^{(5)}(0) = 625, \quad W_0^{(6)}(0) = -7776.$$

Solution to Theorem 15.5. Use the antiderivative $A(x) = xW(x) - x + e^{W(x)}$. On W_0 , the w endpoint runs from -1 to 0 , giving $A(0) - A(-1/e) = 1 - 3/e$. On W_{-1} it runs from -1 to $-\infty$; since $e^w(w^2 - w + 1) \rightarrow 0$ as $w \rightarrow -\infty$, the value is $0 - 3/e = -3/e$. The exponential decay of $dx = e^w(1+w)dw$ dominates the linear growth of w , explaining convergence.

Solution to Theorem 15.6. From (6.2), $p^2 = u^2 + (2/3)u^3 + \mathcal{O}(u^4)$. Seek $u = p + ap^2 + \mathcal{O}(p^3)$. Substitution gives $p^2 = p^2 + (2a + 2/3)p^3 + \mathcal{O}(p^4)$, so $a = -1/3$. Thus $W_0 = -1 + p - p^2/3 + \dots$. The lower branch uses the same analytic inverse at $-p$, hence $W_{-1} = -1 - p - p^2/3 + \mathcal{O}(p^3)$.

Solution to Theorem 15.7. Put $w = L_1 - L_2 + v$. Then

$$0 = v + \log\left(1 - \frac{L_2 - v}{L_1}\right).$$

Since $(L_2 - v)/L_1 \rightarrow 0$, expand the logarithm: $v = (L_2 - v)/L_1 + \mathcal{O}(L_2^2/L_1^2)$. The term v/L_1 is of smaller order, so $v = L_2/L_1 + \mathcal{O}(L_2^2/L_1^2)$.

Solution to Theorem 15.8. Put $u = -W_{-1}(x)$ and $\eta = -\log(-x) = u - \log u$. Since $u \geq 1$, $u \geq \eta$ and therefore $\log u \geq \log \eta$. Hence $u = \eta + \log u \geq \eta + \log \eta$. Multiplication by -1 gives the stated bound.

Solution to Theorem 15.9. Let $w = W_0(x) > 1$. As in the proof of Theorem 10.5, $0 < b_0 < w$. With $r = b_0/w \in (0, 1)$, (10.5) is negative, so $T_x(b_0) < w$. Equation (10.6) is positive because $b_0/(1 + b_0) > 0$, $\log(w/b_0) > 0$, and $w - b_0 > 0$. Thus $b_0 < T_x(b_0) < w$.

Solution to Theorem 15.10. Here $a = -1/4$ and $\tau = 1$, so

$$\lambda_0 = W_0(-1/4), \quad \lambda_{-1} = W_{-1}(-1/4).$$

Both are real because $-1/e < -1/4 < 0$. The principal exponent lies in $(-1, 0)$ and the lower-branch exponent is less than -1 .

Solution to Theorem 15.11. Formula (13.12) gives

$$t_* = -W_{-1}(-e^{-2}) - 2.$$

Since $(-2)e^{-2} < -e^{-2}$ and we^w is decreasing on $(-\infty, -1)$, the solution of $we^w = -e^{-2}$ lies below -2 . Hence $t_* > 0$.

Solution to Theorem 15.12. Writing $y = \log x > 1$ gives $(-y)e^{-y} = -1/n$, so

$$x = e^y = -nW_{-1}(-1/n).$$

The principal value $W_0(-1/n)$ lies in $(-1, 0)$ and would give $0 < y < 1$, contradicting $x > e$.

Solution to Theorem 15.13. The equation $T = ze^T$ has $\phi(t) = e^t$. Lagrange inversion gives

$$[z^n]T^r = \frac{r}{n}[t^{n-r}]e^{nt} = \frac{r}{n} \frac{n^{n-r}}{(n-r)!} = \frac{rn^{n-r-1}}{(n-r)!}.$$

Appendix A

Formal-residue proof of Lagrange–Bürmann inversion

This appendix proves [Theorem 5.1](#) algebraically. No contour or convergence is required.

A.1 Formal residues

For a formal Laurent series $A(t) = \sum_{n \in \mathbb{Z}} a_n t^n$, define

$$\operatorname{Res}_t A(t) dt = a_{-1}.$$

Then

$$[t^n]A(t) = \operatorname{Res}_t A(t) t^{-n-1} dt. \quad (\text{A.1})$$

Also

$$\operatorname{Res}_t \frac{d}{dt} B(t) dt = 0, \quad (\text{A.2})$$

because differentiation can never produce a t^{-1} term: the possible source t^0 has derivative zero.

We need the formal change-of-variable rule. If $z = g(t)$ with $g(t) = g_1 t + g_2 t^2 + \cdots$ and $g_1 \neq 0$, then

$$\operatorname{Res}_z A(z) dz = \operatorname{Res}_t A(g(t)) g'(t) dt. \quad (\text{A.3})$$

It is enough by linearity to check $A(z) = z^m$. If $m \neq -1$, the right side is the residue of $g(t)^m g'(t) = \frac{1}{m+1} (g(t)^{m+1})'$, hence zero by (A.2); the left side is also zero. If $m = -1$, write $g(t) = g_1 t(1 + \mathcal{O}(t))$. Then $g'(t)/g(t) = t^{-1} +$ a power series, so both residues equal 1.

A.2 Proof of the inversion formula

Suppose $w = z\phi(w)$ with $\phi(0) \neq 0$. Then $z = w/\phi(w)$ is a valid formal change of variable. For $n \geq 1$,

$$\begin{aligned} [z^n]H(w(z)) &= \operatorname{Res}_z H(w(z)) z^{-n-1} dz \\ &= \operatorname{Res}_w H(w) \left(\frac{w}{\phi(w)} \right)^{-n-1} d\left(\frac{w}{\phi(w)} \right) \\ &= \operatorname{Res}_w \left[H(w) \phi(w)^n w^{-n-1} - H(w) \phi(w)^{n-1} \phi'(w) w^{-n} \right] dw. \end{aligned} \quad (\text{A.4})$$

Now apply (A.2) to $H(w)\phi(w)^nw^{-n}$:

$$0 = \operatorname{Res}_w [H'(w)\phi(w)^nw^{-n} + nH(w)\phi(w)^{n-1}\phi'(w)w^{-n} \\ - nH(w)\phi(w)^nw^{-n-1}]dw.$$

Rearranging shows that the residue in (A.4) equals

$$\frac{1}{n} \operatorname{Res}_w H'(w)\phi(w)^nw^{-n} dw = \frac{1}{n}[w^{n-1}]H'(w)\phi(w)^n.$$

This is exactly (5.2).

Appendix B

Coefficient tables and recurrences

B.1 Polynomials for higher derivatives

Recall

$$\frac{d^n W}{dx^n} = \frac{e^{-nW} P_n(W)}{(1+W)^{2n-1}}, \quad P_{n+1} = (1+w)P'_n - (nw + 3n - 1)P_n.$$

The first polynomials are

$$\begin{aligned} P_1(w) &= 1, \\ P_2(w) &= -w - 2, \\ P_3(w) &= 2w^2 + 8w + 9, \\ P_4(w) &= -6w^3 - 36w^2 - 79w - 64, \\ P_5(w) &= 24w^4 + 192w^3 + 622w^2 + 974w + 625. \end{aligned}$$

The leading coefficient of P_n is $(-1)^{n-1}(n-1)!$.

Proof. The displayed recurrence was proved in [Theorem 3.3](#); direct substitution gives the table. If the leading coefficient of P_n is a_n and its degree is $n-1$, only the term $-nwP_n$ contributes degree n to P_{n+1} . Hence $a_{n+1} = -na_n$. Starting with $a_1 = 1$ gives $a_n = (-1)^{n-1}(n-1)!$. $\square$

B.2 Puiseux coefficients

For $U(q) = \sum_{n \geq 1} c_n q^n$ in (6.6),

$$c_n = \frac{1}{n} [t^{n-1}] \left(\frac{t^2}{2(1 + (t-1)e^t)} \right)^{n/2}.$$

The first values are

n	c_n	decimal value
1	1	1
2	$-1/3$	-0.3333333333
3	$11/72$	0.1527777778
4	$-43/540$	-0.07962962963
5	$769/17280$	0.04450231481
6	$-221/8505$	-0.02598471487

The W_0 coefficient of p^n is c_n ; the W_{-1} coefficient is $(-1)^n c_n$.

B.3 Logarithmic asymptotic polynomials

Write

$$W_0(x) = L_1 - L_2 + \sum_{\ell \geq 1} \frac{A_\ell(L_2)}{L_1^\ell}.$$

Then

$$A_\ell(y) = \sum_{m=1}^{\ell} \frac{(-1)^{\ell-m}}{m!} \left[\begin{matrix} \ell \\ \ell - m + 1 \end{matrix} \right] y^m. \quad (\text{B.1})$$

The first five are

$$\begin{aligned} A_1(y) &= y, \\ A_2(y) &= \frac{1}{2}y(y-2), \\ A_3(y) &= \frac{1}{6}y(2y^2-9y+6), \\ A_4(y) &= \frac{1}{12}y(3y^3-22y^2+36y-12), \\ A_5(y) &= \frac{1}{60}y(12y^4-125y^3+350y^2-300y+60). \end{aligned}$$

For $-W_{-1}(x)$ with $\eta = \log(1/(-x))$ and $B = \log \eta$, the coefficient of $\eta^{-\ell}$ is

$$B_\ell(y) = \sum_{m=1}^{\ell} \frac{(-1)^{m-1}}{m!} \left[\begin{matrix} \ell \\ \ell - m + 1 \end{matrix} \right] y^m. \quad (\text{B.2})$$

These two families contain the same unsigned Stirling numbers; only the sign pattern changes.

B.4 A compact real-branch table

Branch and input	Output range	Monotonicity	Best elementary local coordinate
$W_0, \quad x \in [-1, 0]$	$[-1, 0]$	increasing	$p = \sqrt{2(1+ex)}$ near $-1/e$; x near 0
$W_0, \quad x \in [0, \infty)$	$[0, \infty)$	increasing	x near 0; $(\log x, \log \log x)$ at infinity
$W_{-1}, \quad x \in [-1/e, 0)$	$(-\infty, -1]$	decreasing	p near $-1/e$; $\eta = \log(1/(-x))$ near 0

Appendix C

Reference implementation outline

The file `lambert_w_experiments.py` included with the article is a fully executable, heavily commented numerical companion. The following pseudocode isolates the evaluator’s essential control flow.

```
function real_lambert_w(x, branch, tolerance):
    reject x < -1/e
    if x == -1/e: return -1

    if branch == 0:
        if x == 0: return 0
        if -1/e < x < 0: b = x
        else if 0 < x < e: b = x/e
        else if x == e: return 1
        else: b = log(x) - log(log(x))

    else if branch == -1:
        reject x >= 0
        s = -1 - log(-x)
        b = -1 - sqrt(2*s) - s

    repeat:
        # Prefer log(abs(x)) - log(abs(b)) at extreme scale.
        log_ratio = log(x/b)
        next = b/(1+b) * (1 + log_ratio)

        if a certified residual or bracket test is below tolerance:
            return next
        b = next
```

Near $-1/e$, initialize from the paired Puiseux expansion rather than from the generic formulas. Near 0^- on W_{-1} , initialize from the real logarithmic asymptotic expansion. These replacements improve speed but are not needed for correctness of the monotone starts.

For arbitrary precision, compute with guard digits and certify the result by an interval or residual bound before rounding. Loczi’s analysis gives explicit global estimates for a closely related implementation [13]; modern complex implementations additionally track branch cuts with ball or interval arithmetic.

Appendix D

Guide to the supplied source archive

The archive mixes foundational references, approximation papers, application papers, and secondary formula collections. The following map records how they informed this article.

Corless–Gonnet–Hare–Jeffrey–Knuth (1996). The canonical modern reference: history, complex branches, asymptotics, numerical evaluation, and symbolic integration. The archive contains both a journal scan and a technical-report rendering [3].

DLMF, Section 4.13. A concise authoritative catalogue of definitions, branch conventions, series, asymptotic formulas, and references [4].

Barry et al. (2000). Separate analytic approximations for W_{-1} , the negative part of W_0 , and the positive part of W_0 , with endpoint matching and iterative use [5].

Banwell (2000). The series-resistance diode and bipolar-transistor formulas used in [Equation \(13.4\)](#), together with circuit-oriented approximations [6].

Golicnik (2012). A review of biochemical kinetic equations whose integrated forms are solved by W , including Michaelis–Menten variants [7].

Chatzigeorgiou (2013). The elementary two-sided W_{-1} bounds proved in [Theorem 8.6](#) and their communication-theory application [8].

Lehtonen (2016). An accessible survey of ecological and evolutionary models, including marginal-value, population, and epidemic equations [9].

Iacono–Boyd (2017). Global principal-branch approximation, the logarithmic Newton map, improved large-argument bounds, and Chebyshev approximation issues [10].

Corcino–Corcino–Mezo (2018). The Fourier–logarithm integral and continued-fraction constructions in [Chapter 9](#) [11].

Visser (2018). The prime-counting reformulation and Lambert-based estimates summarized in [Equation \(13.18\)–\(13.20\)](#) [12].

Loczi (2022). Monotone, quadratically convergent real-branch recursions with explicit uniform high-precision error estimates [13].

Howard (2025) and the accompanying manuscript. Schroder-type inverse series for both real branches, convergence envelopes, and systematic high-order approximations [14].

Secondary formula collections. Useful cross-checks for identities, special values, and symbolic notation. They were treated as secondary compilations; proofs and primary references were preferred in the article.

The archive also includes application-specific and formula-collection PDFs whose examples overlap the themes above. No numerical claim in the article was accepted solely because it appeared in a secondary table: the attached program checks the displayed expansions independently.

Historical note

Johann Heinrich Lambert studied a related transcendental equation in 1758. Leonhard Euler developed the associated series in work published in 1783. The modern notation and systematic branch treatment became standard much later; the 1996 paper of Corless, Gonnet, Hare, Jeffrey, and Knuth played a central role in establishing Lambert W as a standard special function. The name honors Lambert, while the alternative name *product logarithm* reflects that W inverts the product of a variable and its exponential.

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